FLORIDA INTERNATIONAL UNIVERSITY

Miami, Florida

LIGHT DETECTION AND RANGING (LIDAR) BACKPACK DEVELOPMENT FOR AUTOMATED SALT CLOSURE MONITORING AT THE WASTE ISOLATION PILOT PLANT

A thesis submitted in partial fulfillment of the

requirements for the degree of

MASTER OF SCIENCE

in

MECHANICAL ENGINEERING

by

Benjamin Vargas

2026

To: Interim Dean Ines Triay
College of Engineering and Computing

This thesis, written by Benjamin Vargas, and entitled Light Detection and Ranging (LIDAR) Backpack Development for Automated Salt Closure Monitoring at the Waste Isolation Pilot Plant, having been approved in respect to style and intellectual content, is referred to you for judgment.

We have read this thesis and recommend that it be approved.

Dwayne McDaniel

Ibrahim Tansel

Leonel Lagos, Major Professor

Date of Defense: March 27, 2026

The thesis of Benjamin Vargas is approved.

Interim Dean Ines Triay
College of Engineering and Computing

Andrés G. Gil
Senior Vice President for Research and Economic Development
and Dean of the University Graduate School

Florida International University, 2026

© Copyright 2026 by Benjamin Vargas

All rights reserved.

ACKNOWLEDGMENTS

I am deeply grateful to the team at the Florida International University Applied Research Center and the U.S. Department of Energy's Office of Environmental Management – FIU Cooperative Agreement (under Contract No. DE-EM0005213) for their continued support throughout this work. Access to facilities, technical guidance, and daily collaboration made this research possible. I also thank the faculty and staff of FIU's Department of Mechanical and Materials Engineering for providing the academic foundation and mentorship that supported this project. I extend my sincere appreciation to my lab mentor, Mr. Anthony Abrahao, for his consistent guidance, practical insight, and high standard of professionalism. His feedback during system design, integration, and testing strengthened both the technical quality and execution of this work. Finally, I thank the Robotics Lab researchers for their collaboration and shared problem-solving. The collective effort and dedication of the team played an important role in advancing this research and bringing the project to completion.

ABSTRACT OF THE THESIS

LIGHT DETECTION AND RANGING (LIDAR) BACKPACK DEVELOPMENT FOR AUTOMATED SALT CLOSURE MONITORING AT THE WASTE ISOLATION PILOT PLANT

by

Benjamin Vargas

Florida International University, 2026

Miami, Florida

Professor Leonel Lagos, Major Professor

The Waste Isolation Pilot Plant (WIPP) is the United States' only deep geological repository for defense-related transuranic waste and requires continuous monitoring of tunnel deformation caused by salt creep. Traditional instruments provide limited spatial coverage, motivating a mobile, high-fidelity mapping approach. This research develops a wearable Light Detection and Ranging (LIDAR) based system to generate accurate three-dimensional representations of underground corridors.

The system integrates an Ouster OS1 LIDAR, four calibrated cameras, and Simultaneous Localization and Mapping (SLAM) within the Robot Operating System 2 (ROS2). Rigid-body transformations and intrinsic camera modeling project three-dimensional LIDAR points into image space, producing geometrically consistent colorized point clouds. The lightweight frame supports stable, operator-based deployment in GPS-denied environments.

Experimental validation demonstrates metrically consistent reconstructions suitable for temporal alignment and differential analysis. The resulting maps support quantitative change detection to measure salt closure over time, providing a scalable monitoring approach aligned with Department of Energy operational objectives at WIPP.

TABLE OF CONTENTS

CHAPTER PAGE

1. INTRODUCTION 1
1.1 Background 1
1.2 Motivation 4
1.3 WIPP Constraints and Requirements 6

2. LITERATURE REVIEW 10
2.1 Current Mapping Technologies 10
2.1.1 Static Laser Scanning 10
2.1.2 Mobile Laser Scanning 11
2.1.3 3D Reconstruction and Photogrammetry 12
2.1.4 Robotic Platforms for Access-Limited Mapping 12
2.1.5 Use at WIPP and Current Practice 12
2.2 Light Detection and Ranging 14
2.2.1 Pulsed Time-of-Flight Ranging 14
2.2.2 Amplitude-Modulated Continuous-Wave Ranging 14
2.2.3 Frequency-Modulated Continuous-Wave Ranging 15
2.2.4 From Range Samples to 3D Points 15
2.2.5 Return Strength and the LIDAR Equation 16
2.2.6 Supporting Salt-Closure Monitoring 16
2.3 Current LIDAR Technologies 18
2.3.1 Velodyne VLP Series 18
2.3.2 Trimble X-Series 20
2.3.3 Ouster OS-Series 22
2.3.4 Considerations for Underground Monitoring 23
2.4 Simultaneous Localization and Mapping 24
2.4.1 LIDAR-Centric SLAM and Scan Registration 25
2.4.2 Loop Closure and Pose Graph Optimization 26
2.4.3 Relevant Open-Source SLAM Packages 27
2.4.4 Kitware LidarSLAM 28
2.5 LIDAR-Camera Sensor Fusion 28
2.5.1 Geometric Foundations of Fusion 29
2.5.2 Calibration Approaches 30
2.5.3 Colorization and Dense Visual Mapping 31
2.5.4 Benefits and Limitations in Underground Environments 32
2.5.5 Role in Salt Closure Monitoring 32

3. SYSTEM DESIGN . . . 34
3.1 Components Used . . . 34
3.1.1 LIDAR Sensor . . . 34
3.1.2 Cameras . . . 35
3.1.3 Computer for Pointcloud Processing . . . 36
3.1.4 Battery . . . 37
3.2 Generative Designed Mounts . . . 38
3.2.1 Sensor Package Mount . . . 39
3.2.2 Tube Clamp . . . 41
3.2.3 Computer Mount . . . 41
3.2.4 Battery Mount . . . 42
3.3 System Assembly . . . 43
3.4 Sensor Testing . . . 45
3.4.1 LIDAR IP Configuration . . . 45
3.4.2 Bandwidth Restraints with Cameras . . . 45
3.5 ROS2 Integration . . . 46
3.5.1 Ouster Package . . . 46
3.5.2 USB Camera Package . . . 47
3.5.3 LidarSlam Package . . . 47

4. SENSOR FUSION AND COLORIZATION . . . 48
4.1 Transform Architecture and Coordinate Frames . . . 48
4.2 Intrinsic Calibration and Camera Model . . . 52
4.3 Projection-Based Colorization Algorithm . . . 55
4.4 Integration with SLAM and Map Consistency . . . 56
4.5 Validation Assessment and Interpretive Impact . . . 56

5. POINT CLOUD CHANGE DETECTION . . . 60
5.1 Point Cloud Post-Processing . . . 60
5.2 Point Cloud Registration . . . 66

6. CONCLUSION . . . 71
6.1 Summary of Objectives and Contributions . . . 71
6.2 Implications for Salt Closure Monitoring at WIPP . . . 72
6.3 Limitations . . . 73
6.4 Future Work . . . 74
6.4.1 Optimization of the Colorization Node . . . 74
6.4.2 Demonstration at WIPP . . . 76
6.5 Closing Remarks . . . 76

BIBLIOGRAPHY . . . 78

LIST OF FIGURES

FIGURE PAGE

1.1 Overview of the Waste Isolation Pilot Plant [1]. 2

1.2 Shipment of TRU waste [2]. 3

1.3 WIPP undergrounds operations [3]. 3

1.4 Salt convergence at WIPP [4]. 5

1.5 Example of homogeneous surfaces at WIPP [5]. 7

1.6 Robust environment at WIPP [6]. 8

2.1 Manual static measurement in Panel 8's Room 7 at WIPP [7]. 11

2.2 Velodyne Puck Lite LIDAR. 19

2.3 Trimble X7 LIDAR. 21

2.4 Ouster OS1 LIDAR. 22

2.5 SLAM generation example. 24

2.6 Kitware's LidarSLAM algorithm used to create 3D maps. 28

2.7 Point cloud colorization example. 31

3.1 One of four ELP USB cameras. 36

3.2 Jetson Orin Nano computer used for LIDAR backpack system. 37

3.3 Morphie Powerstation Pro battery pack. 38

3.4 CAD model of generatively designed sensor package mount. 40

3.5 Custom tube clamp connecting the sensor mount to the structural frame. 41

3.6 Generatively designed computer mounting bracket. 42

3.7 Battery mounting bracket integrated into the backpack frame. 43

3.8 Full CAD model of the LIDAR backpack. 44

3.9 Workflow of a ROS2 node. 46

4.1 Side view of the LIDAR-camera sensor package with dimensions used for static transform calculations. 49

4.2 Transform (TF) tree of the LIDAR-camera nodal network running on ROS2. 52

4.3 Colorized point cloud generated in FIU's Applied Research Center Robotics Laboratory. 57

5.1 Trimble X7 scanning setup used for controlled millimeter-scale validation. 61

5.2 Cloud-to-cloud distance map showing approximately 2 mm displacement relative to baseline. 63

5.3 Cloud-to-cloud distance map showing approximately 4 mm displacement relative to baseline. 63

5.4 Cloud-to-cloud distance map showing approximately 6 mm displacement relative to baseline. 64

5.5 Cloud-to-cloud distance map showing approximately 8 mm displacement relative to baseline. 64

5.6 Cloud-to-cloud distance map showing approximately 10 mm displacement relative to baseline. 65

5.7 Overview of the Port of Miami Tunnel in Miami, FL. 67

5.8 Port of Miami tunnel reconstruction shown in two registered halves prior to final stitching. 68

5.9 Full 171 m section of the Port of Miami tunnel stitched into a single continuous point cloud using SLAM-based registration. 68

LIST OF ABBREVIATIONS

Abbreviation	Meaning
WIPP	Waste Isolation Pilot Plant
FIU	Florida International University
DOE	Department of Energy
TRU	Transuranic
LIDAR	Light Detection and Ranging
SLAM	Simultaneous Localization and Mapping
ROS2	Robot Operating System 2
IMU	Inertial Measurement Unit
LOAM	LIDAR Odometry and Mapping
ICP	Iterative Closest Point
C2C	Cloud-to-Cloud
C2M	Cloud-to-Mesh
GPU	Graphics Processing Unit
CUDA	Compute Unified Device Architecture
RGB	Red, Green, Blue
TF	Transform Frame

ARC	Applied Research Center
CAD	Computer-Aided Design
USB	Universal Serial Bus

LIST OF SYMBOLS

Symbol	Meaning
x	Horizontal coordinate in the map frame
y	Vertical coordinate in the map frame
z	Height coordinate
θ	Orientation angle
R	Rotation matrix
t	Translation vector
T	Homogeneous transformation matrix
$\mathbf{p}$	3D point in space
$\mathbf{p}'$	Transformed 3D point
p_i	i-th point in a point cloud
q_i	Corresponding point in a second point cloud
d_i	Minimum Euclidean distance between points
M	Point cloud map
$M(t)$	Point cloud map at time t
$d(\cdot)$	Distance metric between point clouds or surfaces
u	Pixel coordinate (horizontal)

v	Pixel coordinate (vertical)
f_x	Focal length in x-direction
f_y	Focal length in y-direction
c_x	Principal point x-coordinate
c_y	Principal point y-coordinate
K	Camera intrinsic matrix
D	Distortion coefficients
n	Surface normal vector
α	Angle between vectors
Δt	Time difference between sensor measurements
σ_s	Measurement noise standard deviation
σ_r	Registration uncertainty standard deviation
σ_d	Effective deformation detection threshold
$\Delta(x)$	Deformation field between registered maps
$\|\cdot\|$	Euclidean norm

CHAPTER 1

INTRODUCTION

1.1 Background

The enduring management of radioactive waste constitutes one of the most persistent challenges linked to nuclear technology. In the United States, substantial amounts of radioactive waste were produced during World War II and the Cold War due to nuclear weapons research, development, and production initiatives. These efforts led to the generation of transuranic (TRU) waste, principally composed of materials contaminated with long-lived alpha-emitting radionuclides, including plutonium [8]. The primary goal of the U.S. Department of Energy (DOE) has been to guarantee the secure separation of nuclear waste from the biosphere across geological durations.

The Waste Isolation Pilot Plant (WIPP), situated in Carlsbad, New Mexico, was created to tackle this issue and functions as the nation's sole deep geological repository for defense-related transuranic waste. Sanctioned by Congress via the Waste Isolation Pilot Plant Land Withdrawal Act, WIPP commenced disposal operations in 1999 and receives waste produced by DOE facilities around the United States [3]. The facility is essential to the national nuclear cleanup initiative by offering a permanent disposal solution that mitigates environmental hazards and long-term management obligations.

WIPP is situated roughly 650 meters beneath the surface in the Salado Formation, a substantial salt deposit chosen for its advantageous geomechanical and hydrological characteristics (as shown in Figure 1.1). Rock salt demonstrates minimal permeability, lacks moving groundwater, and undergoes plastic deformation under prolonged stress. This plastic deformation induces the progressive creep of

salt, leading to the closure of cracks, voids, and excavated apertures, a phenomenon generally known as natural closure [9]. As disposal chambers undergo deformation over time, the adjacent salt encases the waste, offering a passive and self-sealing isolation mechanism independent of active engineered systems.



Figure 1.1: Overview of the Waste Isolation Pilot Plant [1].

The long-term safety approach at WIPP depends on the synergistic performance of manmade barriers and natural geological processes, as illustrated in Figure 1.2. Waste containers, backfill materials, and repository design elements are enhanced by the inherent creep characteristics of salt, which gradually diminishes empty space and constrains possible routes for radioactive migration [10]. Performance evaluations conducted by the Department of Energy and assessed by the U.S. Environmental Protection Agency, as shown in Figure 1.3, analyze repository functionality over regulatory durations exceeding 10,000 years.



Figure 1.2: Shipment of TRU waste [2].



Figure 1.3: WIPP undergrounds operations [3].

Besides its disposal function, WIPP possesses national importance as a singular operational demonstration of deep geological disposal for radioactive waste in the United States. The facility facilitates remediation initiatives at several historical DOE locations and supplies essential data for comprehending long-term repository

dynamics. Ongoing surveillance of subterranean conditions, particularly tunnel deformation due to salt creep, is crucial for maintaining operating safety, adhering to regulations, and verifying long-term performance.

The subterranean conditions of WIPP impose distinctive operational limitations, such as restricted access, changing tunnel configurations, and diminished vision. These reasons drive the advancement of sophisticated sensing and mapping systems that can precisely delineate subsurface geometry in areas devoid of GPS. High-fidelity spatial data are essential for quantifying deformation trends, facilitating maintenance planning, and improving situational awareness, hence directly motivating the development of the Light Detection and Ranging (LIDAR)-based mapping system in this research.

1.2 Motivation

Comprehending and measuring the rate of salt creep is essential for the secure operation and long-term performance evaluation of WIPP. In salt formations, the ongoing viscoplastic deformation of halite dictates the sealing of excavated voids, the development of porosity and permeability, and finally the entrapment of deposited waste [11].

Salt creep affects the design and predictive models employed in performance evaluations by regulating the rate of shrinkage in mined areas and the healing of fractures. Preliminary studies at WIPP revealed that the measured closure rates underground were markedly more rapid than initial laboratory predictions, prompting an ongoing research initiative to align field observations with laboratory results and enhance constitutive models of salt deformation [11]. Precise characterization of

creep rate is essential to guarantee that models accurately predict repository closure behavior throughout operational and regulatory timeframes.

Operationally, salt creep impacts ground control in subterranean settings. The gradual yet continuous deformation of the ceiling, ribs, and floor of access drifts necessitates regular maintenance, fastening, and monitoring to provide safe working conditions and safeguard installed infrastructure [12]. Should creep rates surpass projections, ground support systems must adjust to mitigate any undue deformation that may endanger workers or equipment.



Figure 1.4: Salt convergence at WIPP [4].

The capacity of salt to flow and self-seal is fundamental to the safety argument of WIPP from a performance assessment standpoint. Salt creep gradually diminishes void space and compacts crushed salt backfill, steering the repository towards a low-permeability condition that restricts fluid travel paths [13]. Creep-induced closure, as seen in Figure 1.4, directly affects essential repository performance indicators,

such as brine transport, alterations in disturbed rock zone characteristics, and the development of permeability surrounding excavations [14].

Measuring salt creep rates also guides engineering decisions about trash emplacement sequencing and retrievability. Accelerated closure rates may need more frequent scheduling of garbage insertion and backfilling activities to maintain accessibility of apertures during operations. Conversely, comprehending long-term creep behavior facilitates the planning for post-closure performance and regulatory-required predictions of radioactive containment over millennia. These considerations collectively designate the assessment of salt creep rates as a pragmatic and scientific priority for WIPP. Precise creep metrics enhance model accuracy, bolster operational safety, and strengthen the enduring isolation properties essential for the repository's regulatory adherence and national objectives.

1.3 WIPP Constraints and Requirements

The subterranean conditions at WIPP impose a series of physical, operational, and legal limitations that directly affect the viability of deformation monitoring equipment. Access to disposal panels and service drifts is constrained both spatially and temporally because of ongoing waste emplacement, ground control operations, and safety regulations regarding worker exposure [3]. As salt creep gradually modifies tunnel geometry, monitoring systems must function inside constantly evolving cross-sections and in restricted regions where fixed instrumentation coverage is necessarily constrained.

Visibility conditions in the subterranean environment are difficult and fluctuate with operational activities. The illumination is artificial and uneven, dust may accumulate from vehicular activity or maintenance tasks, and surfaces frequently

appear optically uniform, as illustrated in Figure 1.5. These variables diminish the efficacy of conventional optical surveying methods and hinder feature-based measures that depend on significant visual contrast. The subterranean environment is entirely devoid of GPS, necessitating the use of local coordinate systems and internally coherent localization techniques for all spatial data references.



Figure 1.5: Example of homogeneous surfaces at WIPP [5].

From an operational perspective, monitoring technology must reduce disruption to standard repository operations. Stationary convergence monitoring devices, however efficient at specific sites, offer restricted spatial resolution and necessitate installation, upkeep, and regular recalibration [12]. These constraints elevate labor needs and constrain the capacity to swiftly characterize deformation throughout extensive tunnel networks. Consequently, there is a want for mobile sensing technologies that can obtain spatially continuous measurements without the need for permanent installation.

Technologies utilized underground must be resilient to severe environmental conditions as illustrated in Figure 1.6. Equipment must withstand high humidity, salt dust, vibrations from vehicular traffic, and prolonged operation in confined environments. Power availability is limited, encouraging devices with minimal energy use or battery-operated functionality. Data acquisition systems must function independently due to the constraints of wireless transmission capacity and the assumption of continuous distant access being unreliable.



Figure 1.6: Robust environment at WIPP [6].

To adequately facilitate salt creep monitoring at WIPP, a sensing platform must fulfill certain essential criteria. The system must deliver centimeter-level geometric precision across extended corridor distances to identify gradual deformation trends. It must operate dependably in GPS-denied contexts and sustain localization consistency despite restricted geometric features. The platform must be portable and swiftly deployable to decrease operating overhead and mitigate disruption to subterranean activities. The system must produce consistent, high-fidelity spatial datasets

that facilitate quantitative comparisons over time, allowing for the identification of subtle convergence rates rather than only qualitative observations.

These constraints jointly drive the creation of a transportable, LIDAR-based mapping device proficient in generating dense three-dimensional representations of subterranean geometry. This technology enhances current fixed equipment by offering geographical context, broadening coverage over entire drifts and panels, and facilitating deformation evaluation at a scale aligned with the operational and safety requirements of WIPP.

CHAPTER 2

LITERATURE REVIEW

2.1 Current Mapping Technologies

Subterranean facilities are increasingly relying on 3D spatial data for inspection, documentation, and change detection in areas without GPS. Today's technologies at WIPP include (1) static laser scanning and classic surveying procedures, (2) mobile scanning equipment for walk-through or vehicle data collection, and (3) image-based reconstruction methods for building 3D models from photographic data. The primary purpose of these methods is to offer metrically consistent geometry that enables temporal comparison and quantification of deformation, clearance changes, and infrastructure status.

2.1.1 Static Laser Scanning

Terrestrial static laser scanning is a popular reference technique for producing high-precision as-built documentation in tunnels, drifts, and other constrained environments; however, it often necessitates multiple scanning installations, line-of-sight considerations, and time-consuming field operations. This minimizes the frequency of static searches in big networks, particularly when access times are limited or neighboring activity must continue. Figure 2.1 demonstrates that static scanning may achieve low centimeter-level errors under controlled procedures, making it suitable for baseline models and validation datasets [15].



Figure 2.1: Manual static measurement in Panel 8's Room 7 at WIPP [7].

2.1.2 Mobile Laser Scanning

Mobile laser scanning has emerged as the preferable solution in cases where coverage length and field duration are critical operating restrictions. Recent tunnel studies show that mobile scanning methods achieve equivalent accuracy at tunnel size to static procedures while significantly increasing measurement speed, enabling for more frequent mapping of vast corridors [15]. In deformation-monitoring scenarios, repeated mobile scans are often regarded as an excellent method for transitioning from sparse, point-based observations to spatially continuous geometry that includes the entire drift length. Current tunnel deformation research demonstrates a move from manual or constrained techniques to mobile point-cloud collecting for change detection in vast tunnel networks [16].

2.1.3 3D Reconstruction and Photogrammetry

Photogrammetry and image-based reconstruction techniques are used to create 3D representations of subterranean environments for research and documentation. These methods can produce visually interpretable results while using relatively lightweight camera equipment. Tunnel-centric research shows that expedited capture workflows can produce models that align with laser scan references at the centimeter scale for brief segments, achieving significant reductions in acquisition time compared to traditional imaging methods [17]. Image-based techniques are typically utilized when visual context is crucial, payload weight must be reduced, or the operational goal prioritizes qualitative condition evaluation with geometry.

2.1.4 Robotic Platforms for Access-Limited Mapping

The use of robotic platforms to reduce worker exposure and access sites that are difficult, unstable, or otherwise unsuitable for direct entry has been a big step forward in subsurface mapping. In these circumstances, the mapping goal is often linked to situational awareness and early examination, followed by more precise modeling after access is restored. This operating pattern is compatible with the widespread usage of mobile sensing payloads on unmanned devices to provide rapid assessment of limited subsurface areas prior to deploying troops for entry.

2.1.5 Use at WIPP and Current Practice

Geotechnical monitoring at WIPP is crucial for ensuring operational safety and determining long-term performance. WIPP's geotechnical program plan specifies an instrumentation-based monitoring methodology that includes tape extensometers, convergence meters, borehole extensometers, piezometers, strain gauges, load cells,

and crack meters [18]. This methodology enables regular assessment of excavation stability and closure, resulting in periodic datasets that highlight potential instability zones and inform correction processes [18].

According to the regulatory and compliance monitoring documents, this activity is a systematic monitoring program with established parameters, collection frequencies, and annual reports. The 2019 Compliance Recertification Application appendix defines geotechnical monitoring metrics such as creep closure, stress measurements, and brittle deformation indicators. Data collection frequency ranges from weekly to monthly, depending on repository circumstances, with annual reporting and assessment cycles [19]. According to the article [19], the monitoring system includes both remotely monitored and manually read devices. Engineers collect data quarterly or as frequently as needed.

In addition to these well-known procedures, WIPP has recently utilized robotic inspection to facilitate assessments in restricted access areas. The DOE has announced a 2024 subsurface aerial survey in which an unmanned aerial system would visually assess an underground exhaust drift that was previously only accessible to personnel. This was accomplished using onboard video cameras and LIDAR to allow for remote examination before determining the suitability of human ingress [20]. This example demonstrates a current operational framework in which remote sensing is used to aid in safety decisions, followed by higher-resolution assessments once conditions have been evaluated.

WIPP relies on point-based instrumentation and inspections but lacks the ability to rapidly generate continuous 3D geometry across large drift networks. Mobile mapping addresses this gap by providing consistent, corridor-scale data for quantitative change analysis while meeting WIPP's operational and safety constraints.

2.2 Light Detection and Ranging

LIDAR estimates distance by aiming laser energy at a surface and detecting the reflected signal. A 3D scan is made by continuously measuring the range from multiple directions while tracking the sensor's position, resulting in a collection of points with coordinates and properties (such as return intensity). The measurement ideas are the same across all ranging modalities, but the encoding of time, phase, or frequency information vary between broadcast and received signals [21, 22].

2.2.1 Pulsed Time-of-Flight Ranging

In pulsed time-of-flight ranging, the device sends out a brief optical pulse and measures the round-trip time between transmission and reception. The line-of-sight distance R is calculated by using the measured round-trip duration Δt:

$$R = \frac{c\,\Delta t}{2}, \tag{2.1}$$

where c is the speed of light [21]. The factor of $1/2$ appears because the pulse travels to the target and back. One practical application of 3D scanning is that the precision of time measurement determines the available range resolution. With a time uncertainty of σ_t, first-order error propagation yields

$$\sigma_R \approx \frac{c}{2}\,\sigma_t. \tag{2.2}$$

This relationship leads to designs that minimize timing jitter and enhance return detection stability in low-reflectivity or dusty situations [23].

2.2.2 Amplitude-Modulated Continuous-Wave Ranging

In amplitude-modulated continuous-wave (AMCW) ranging, the emitted optical radiation is modulated at frequency f_m, and the receiver calculates the phase delay

$\Delta\phi$ between the sent and received modulation. Estimated range is

$$R = \frac{c}{4\pi f_m}\,\Delta\phi, \tag{2.3}$$

with $\Delta\phi$ typically wrapped to $[0, 2\pi)$, which introduces an ambiguity interval

$$R_{\text{amb}} = \frac{c}{2f_m}. \tag{2.4}$$

AMCW is frequently employed alongside time-of-flight techniques to measure depth based on temporal aspects of the return signal [24, 25].

2.2.3 Frequency-Modulated Continuous-Wave Ranging

In frequency-modulated continuous-wave (FMCW) ranging, the optical frequency chirps. The incoming signal is combined with a local reference to provide a beat frequency f_b, which represents the delay. For a linear chirp with a slope $S = B/T$ (where B is bandwidth and T is sweep time), a frequent connection is

$$R = \frac{c}{2S}\,f_b. \tag{2.5}$$

A key result is that range resolution scales with bandwidth:

$$\Delta R \approx \frac{c}{2B}. \tag{2.6}$$

FMCW offers a coherent alternative to pulsed and AMCW techniques, with measurements based on frequency content after mixing [24].

2.2.4 From Range Samples to 3D Points

Each measurement yields a range R_i along a known beam direction. In the LIDAR coordinate frame, a common point parameterization is

$$\mathbf{p}_i^L = R_i\,\hat{\mathbf{u}}_i, \tag{2.7}$$

where $\hat{\mathbf{u}}_i$ is a unit vector for the beam direction (determined by scan angles). To express points in a fixed map frame, the system applies the estimated pose (rotation $\mathbf{R}$ and translation $\mathbf{t}$) at the acquisition time:

$$\mathbf{p}_i^M = \mathbf{R}_{ML}\,\mathbf{p}_i^L + \mathbf{t}_{ML}. \tag{2.8}$$

This transformation transforms unprocessed ranging into a metrically coherent point cloud appropriate for mapping and temporal comparison [21, 22].

2.2.5 Return Strength and the LIDAR Equation

The amount of optical power provided to the receiver once it engages with the target and transmits over the channel determines the detectability of the return. The LIDAR range equation [26, 27] shows that received power decreases with distance (typically with an inverse-square relationship) and is affected by transmitter power, receiver aperture, target reflectance or backscatter, and transmission losses. This paradigm describes how dust, surface features, and incidence angle affect data density and noise patterns during a scan in subterranean environments.

2.2.6 Supporting Salt-Closure Monitoring

In salt-closure monitoring, a consistent three-dimensional depiction of drift geometry is more important than a single range measurement. The aforementioned equations are linked to the monitoring workflow in three explicit ways:

1) Metric geometry for change detection. Equation 2.1 (or its AMCW/FMCW equivalents) defines the radial distances that make up the point cloud. Equations 2.7–2.8 illustrate how distances are converted into three-dimensional coordinates inside a stable reference frame. This enables the comparison of scans from different

dates using cloud-to-cloud distance, cross-sectional profiles, and surface-to-surface deviation maps. The monitoring problem becomes a geometric differencing challenge rather than a sparse point-instrument conundrum [21].

2) Budgetary uncertainty for modest deformations. Equation 2.2 illustrates a first-order link between timing uncertainty and range uncertainty. In conjunction with registration and pose imperfections, it enables an error budget for the lowest visible convergence rate over a certain time period. This is crucial for setting the frequency of follow-up surveys and establishing acceptable values for change maps.

3) Operational stability in low-contrast, dusty environments. The LIDAR equation shows how data completeness varies with range and visibility while the underlying geometry remains constant. This encourages collecting procedures that increase repeatability (uniform traversal paths, constant scan coverage, and quality judgments on return density) and post-processing processes that accommodate for fluctuating return strength [26].

These findings lend support to LIDAR as a viable measuring foundation for spatially continuous deformation monitoring in a repository environment without GPS. They also present specific variables that can be linked to system-level criteria (detectable deformation magnitude, repeat mapping interval, permitted drift, and scan coverage limitations) before getting into specific sensor hardware.

2.3 Current LIDAR Technologies

Commercial LIDAR systems differ greatly in architecture, scanning methodology, data transfer rates, range capabilities, and target application areas. The most significant variations in subsurface deformation monitoring are (1) static versus mobile operation, (2) mechanical versus solid-state scanning, (3) data density versus portability, and (4) integration flexibility with robotic or wearable platforms. This section looks at three representative system families that are important to subsurface mapping and corridor-scale documentation.

2.3.1 Velodyne VLP Series

The Velodyne VLP series (Figure 2.2) consists of mechanically rotating, multi-channel LIDAR sensors, originally designed for autonomous vehicle research and robotics applications [28]. These devices give 360-degree horizontal coverage by continuously rotating, while vertically stacking multiple laser channels to provide a three-dimensional field of vision.



Figure 2.2: Velodyne Puck Lite LIDAR.

The VLP architecture enables high data rates while remaining lightweight, making it appropriate for mobile robotics and real-time mapping applications. Conventional systems in this category have scan speeds ranging from 5 to 20 Hz and create hundreds of thousands of points per second, allowing for the creation of thick point clouds during continuous movement. Their small size and low power consumption make them excellent for use in wearable and terrestrial robotic equipment.

Mechanically rotating multi-beam devices in subterranean environments enable rapid corridor coverage and are compatible with Simultaneous Localization and Mapping (SLAM)-based localization. Nonetheless, performance is determined by return signal intensity, incidence angle, and ambient particles such as dust. Although these sensors are not as exact as tripod-mounted terrestrial scanners, their mobility allows for repeated re-surveys of huge drifts, which is crucial for tracking gradual deformation trends rather than isolated absolute readings.

2.3.2 Trimble X-Series

The Trimble X-Series is high-precision terrestrial laser scanning equipment intended for static survey applications and depicted in Figure 2.3 [29]. These sensors produce dense, high-precision point clouds by combining phase-based or time-of-flight ranging with high-angular-resolution scanning techniques.

Unlike mobile rotating devices, terrestrial scanners in this category are tripod-mounted and collect data from fixed locations. They are widely utilized in construction documentation, industrial metrology, and geographic survey applications that require millimeter-to-centimeter accuracy within predefined scan volumes. Instead of continuously rotating the sensor body, internal mirrors or scanning assemblies revolve to offer full 360-degree coverage.



Figure 2.3: Trimble X7 LIDAR.

Terrestrial scanners provide detailed baseline geometry for subterranean monitoring and act as reference datasets for testing mobile mapping systems. Their constraints stem from operational overhead. Several scanning configurations are required for lengthy drifts, and repositioning lengthens survey duration and personnel requirements. In places like WIPP, where access windows are limited and ground conditions change often, repeating full-tripod campaigns may not be operationally efficient for regular monitoring cycles.

2.3.3 Ouster OS-Series

The Ouster OS-Series, as shown in Figure 2.4, is a series of digital LIDAR sensors designed with solid-state beam-producing and digital signal processing frameworks [30]. These sensors provide multi-beam vertical resolution and large 360-degree horizontal coverage through rotating scanning, comparable to classic spinning sensors, but with a fully digital design that facilitates synchronization, configurability, and integration flexibility.



Figure 2.4: Ouster OS1 LIDAR.

One distinguishing feature of digital LIDAR systems is their ability to communicate correctly timestamped point clouds and other data streams via Ethernet connection protocols. This enables integration with robotic middleware and multisensor systems like camera arrays and inertial measurement units. Changeable angular resolution and programmable data rates enable spatial density optimization within bandwidth limits.

Digital spinning LIDAR systems provide a balance of mobility and geometric accuracy for subsurface applications. Their compatibility with robotic operating systems allows for SLAM-based localization in GPS-denied settings. When deployed in wearable or robotic forms, such systems can explore wide corridors on a constant basis, allowing enabling corridor-scale change detection over time while remaining within power and weight limits.

2.3.4 Considerations for Underground Monitoring

There are trade-offs within these system categories that directly impact their applicability for salt-closure monitoring:

- *Precision versus mobility:* Tripod-mounted terrestrial scanners have superior geometric accuracy, but they require static configurations and extended acquisition durations. Mobile spinning devices forgo absolute precision in order to provide quick spatial coverage.
- *Data density versus operational burden:* High-density static scans produce detailed surface models, but they may not be suitable for frequent full-drift repetition. Mobile systems offer regular mapping at shorter intervals.
- *Integration flexibility:* Sensors developed for robotic integration can be linked with cameras and inertial systems, allowing for continuous mapping in GPS-denied situations.

The operational goal of WIPP-scale drift monitoring is to achieve high single-epoch precision while maintaining constant corridor-scale geometry under constrained access situations. As a result, the selection of a system is based on a combination of geometric precision, deployment efficacy, environmental resilience, and adherence

to localization standards. These factors necessitate an assessment of mobile digital LIDAR designs in comparison to wearable and robotic mapping systems.

2.4 Simultaneous Localization and Mapping

When GPS is not available, SLAM computes the sensor trajectory and generates a coherent 3D map. The most typical failure mechanism in subterranean tunnels is the accumulation of drift caused by minute frame-to-frame changes. Loop closure and global optimization (shown in Figure 2.5) minimize drift when the platform encounters previously mapped regions, which is crucial for repeat mapping for temporal change detection [31, 32].

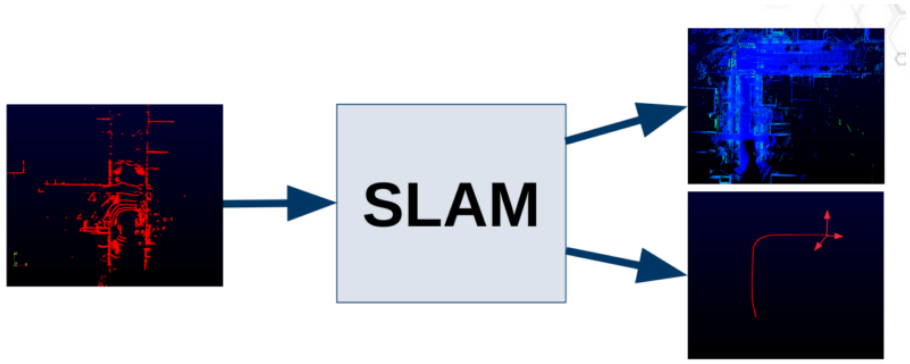


Figure 2.5: SLAM generation example.

In terms of corridor-scale deformation monitoring, SLAM should be considered a facilitating layer. It generates an estimated trajectory, a registered point cloud map, and intermediate states, which may be exported and compared between surveys. This allows for repeatable corridor-scale datasets, enhancing point-based instrumentation with spatially continuous geometry for differencing and cross-section analysis [16].

2.4.1 LIDAR-Centric SLAM and Scan Registration

LIDAR-centric SLAM serves as the geometric foundation for this effort, as precise map-to-map comparison is based on scan registration and trajectory estimation accuracy. In SLAM, incremental motion is frequently studied using scan registration, which aligns each fresh point cloud with a local map or a previous scan. A widely used formulation shortens the distance metric between corresponding sets of points. In its fundamental version, Iterative Closest Point (ICP) addresses

$$\min_{\mathbf{R},\mathbf{t}} \sum_{i} \|\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i\|^2 , \tag{2.9}$$

where $\mathbf{p}_i$ and $\mathbf{q}_i$ are matching points, and $(\mathbf{R}, \mathbf{t})$ is the rigid transformation between scans. Variants increase robustness by employing voxel downsampling, keyframing, normal-based matching, and alignment with a continuously updated local submap rather than a single frame.

Local registration may maintain numerical stability in lengthy, visually repeated tunnels like those at WIPP, as minor alignment errors accumulate over time. This accumulation results in global drift, which means that the reconstructed geometry eventually deviates from its exact configuration, despite steady local scan-to-scan alignment. This drift is crucial for deformation monitoring. If left unchecked, drift can obscure genuine geometric changes or erroneously suggest deformation when none exists.

Loop closure and posture graph modification are utilized to mitigate the effects of trajectory overlap [31, 32]. In pose graph formulation, each estimated sensor pose is represented as a node, with constraints representing the relative transformations between poses. When returning to a previously mapped place, the platform adds a

new constraint and globally optimizes the entire path. This distributes accumulated inaccuracy over the map, reducing substantial distortion.

The purpose of this thesis is not only to make visually consistent maps but also to allow for quantitative comparisons of two surveys conducted on different dates. Assume $\mathcal{M}_1$ and $\mathcal{M}_2$ are point cloud maps acquired at times t_1 and t_2. Following registration, the deformation field is examined using geometric differencing:

$$\Delta(\mathbf{x}) = d\big(\mathcal{M}_1(\mathbf{x}), \mathcal{M}_2(\mathbf{x})\big), \tag{2.10}$$

where $d(\cdot)$ represents a cloud-to-cloud or surface-to-surface distance metric. The reliability of $\Delta(\mathbf{x})$ depends directly on SLAM consistency within each individual survey and on accurate global alignment between surveys.

As a result, LIDAR-centric SLAM in this study performs three critical functions: (1) producing metrically consistent corridor-scale geometry for each survey, (2) reducing drift to maintain accurate spatial relationships over long distances, and (3) enabling repeatable coordinate frames for cross-epoch registration. In the absence of proper scan registration and drift control, computational artifacts would have a bigger impact on deformation estimations than actual physical salt closure. As a result, SLAM is considered as a precision-enhancing element that adds to the primary purpose of detecting map-to-map changes rather than a visualization tool.

2.4.2 Loop Closure and Pose Graph Optimization

Pose graph optimization represents postures as nodes and restrictions as edges. Incorporating a loop closure edge results in a global limitation that corrects drift by optimizing the entire route and then updating the map. The Kitware process

includes a pose graph that can be optimized with loop closure limitations [**?**], a widely used method in many systems.

2.4.3 Relevant Open-Source SLAM Packages

Several open-source packages are commonly used for LIDAR-based mapping:

Cartographer: A widely popular system that integrates local SLAM, global optimization, and loop closure. It was first utilized in 2D but has now been extended to 3D processes [31].

LOAM family: Comprises feature-based LIDAR odometry and mapping (LOAM), which has a significant impact on future pipelines. Its versions aim for real-time performance and support different terrain assumptions [33, 34].

LIDAR-inertial methods: LIDAR-inertial approaches are systems that integrate Inertial Measurement Units (IMUs) and LIDAR to improve motion predictions in the face of difficult dynamics or geometric constraints. Some notable examples are smoothing and filtering techniques [35, 36].

ICP-centric odometry: Recent work highlights the importance of simplicity and robustness in ICP-centric odometry, utilizing contemporary ICP-style registration, rigorous preprocessing, and motion management [37].

Package selection is based on mission-relevant restrictions. WIPP-style corridors include the following characteristics: consistency across extended distances, robustness to uniform geometry, and the ability to export intermediate states for future investigation and comparison.

2.4.4 Kitware LidarSLAM

This study employs Kitware's LidarSLAM tool (Figure 2.6), which enables practical mapping operations that are compatible with corridor-scale capture and post-processing. Kitware offers a ROS2 wrapper and a LidarView-integrated workflow for performing SLAM, adjusting settings, and exporting findings for further analysis [38, 39, 40]. Kitware optimizes graphs and closes loops in the LidarView toolchain to minimize drift during revisits or overlaps [32].

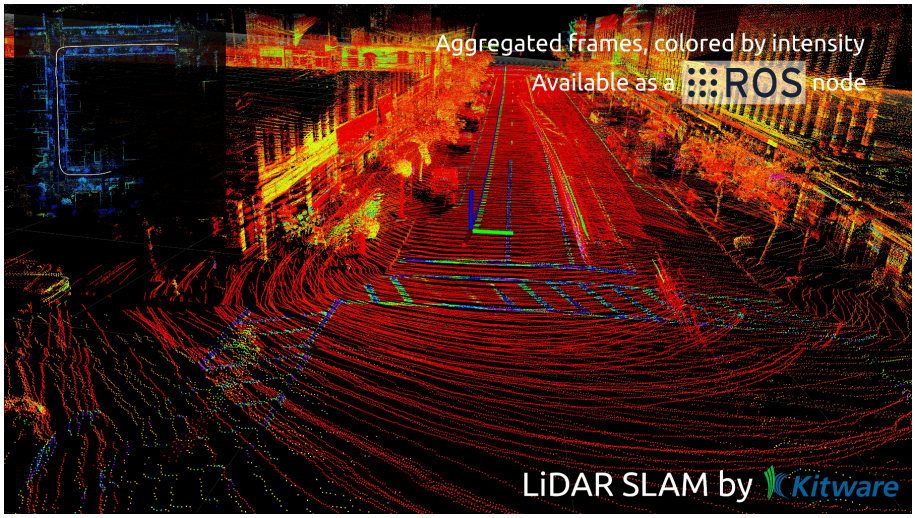


Figure 2.6: Kitware's LidarSLAM algorithm used to create 3D maps.

This thesis employs SLAM as the primary framework for trajectory and registration, yielding metrically consistent point clouds suited for change detection. Instead of a single display model, the goal is to develop repeatable 3D geometry that enables deformation pattern comparison across surveys.

2.5 LIDAR-Camera Sensor Fusion

LIDAR and camera systems offer complementary sensing capabilities. LIDAR uses active ranging to directly measure metric depth, resulting in three-dimensional point

clouds that are geometrically correct. Cameras gather passive radiometric data, which gives detailed color and texture information but lacks intrinsic depth without multi-view reconstruction. Sensor fusion integrates many modalities to create spatially coherent geometry enhanced with visual information.

This fusion is especially effective in subterranean environments. LIDAR maintains geometric coherence in GPS-denied areas, whereas cameras improve interpretability by identifying material limits, structural features, and surface conditions that geometry alone cannot reveal. The integrated system makes it easier to track quantitative deformation and do qualitative inspections.

2.5.1 Geometric Foundations of Fusion

The integration of LIDAR and cameras is based on recognized geometric correlations between sensor coordinate systems. Consider a three-dimensional point represented by the LIDAR coordinate system:

$$\mathbf{p}^L = \begin{bmatrix} x_L \\ y_L \\ z_L \end{bmatrix}. \tag{2.11}$$

To relate this point to the image pixels, it must first be translated to the camera coordinate frame using extrinsic calibration parameters:

$$\mathbf{p}^C = \mathbf{R}_{CL}\mathbf{p}^L + \mathbf{t}_{CL}, \tag{2.12}$$

where $\mathbf{R}_{CL}$ is the rotation matrix and $\mathbf{t}_{CL}$ is the translation vector between the LIDAR and camera frames. The transformed point is then projected into the image plane using the pinhole camera model:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \mathbf{K} \begin{bmatrix} x_C/z_C \\ y_C/z_C \\ 1 \end{bmatrix}, \tag{2.13}$$

where $\mathbf{K}$ is the intrinsic camera matrix, and (u, v) are the image's pixel coordinates. This projection model offers the mathematical foundation for colorizing point clouds, as each 3D point may be assigned RGB values derived from the corresponding image pixel. Accurate fusion requires three calibrated components:

- Intrinsic camera parameters,
- Extrinsic transformation between sensors,
- Temporal synchronization between data streams.

Any of these errors immediately result in color misalignment or ghosting artifacts in the fused point cloud.

2.5.2 Calibration Approaches

Extrinsic calibration between LIDAR and cameras can be done with calibration targets visible in both modalities or using structure-based alignment methods. Target-based techniques frequently employ checkerboard or planar patterns to assess relative posture while minimizing reprojection error. Structure-based approaches make use of natural scene geometry to optimize the stiff transformation that aligns LIDAR and image features. The calibration issue is typically described as a nonlinear optimization problem:

$$\min_{\mathbf{R},\mathbf{t}} \sum_i \|\mathbf{u}_i - \pi(\mathbf{R}\mathbf{p}_i^L + \mathbf{t})\|^2, \tag{2.14}$$

where $\pi(\cdot)$ represents the projection function and $\mathbf{u}_i$ are observed pixel coordinates. Reducing reprojection error guarantees geometric homogeneity across all modalities.

2.5.3 Colorization and Dense Visual Mapping

Colorization is the most simple and straightforward method of merging LIDAR and cameras. Following projection, each 3D point receives an RGB value derived from the image. When run across numerous frames, this results in a colorized point cloud that blends metric geometry and radiometric texture and is depicted in Figure 2.7.

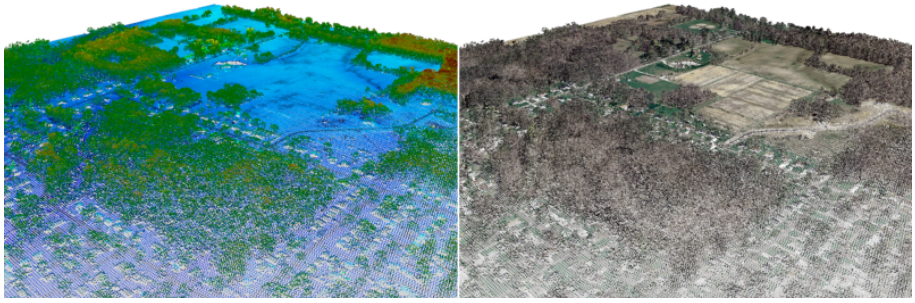


Figure 2.7: Point cloud colorization example.

Advanced fusion approaches build on this concept by incorporating depth and color throughout the mapping process. Certain SLAM methods conserve voxel grids or surfel maps, which maximize both geometry and color concurrently. Others include LIDAR and camera characteristics into tightly integrated state prediction systems. These techniques increase localization robustness and map integrity, but they require greater processing power and strict synchronization.

The fundamental goal of subsurface monitoring is not real-time dense visual SLAM, but rather the acquisition of repeatable, metrically consistent geometry rich in interpretable color. A projection-based colorization pipeline is appropriate and

advantageous because of its versatility. LIDAR SLAM technology creates a stable geometric framework, and camera imagery improves the map during post-processing.

2.5.4 Benefits and Limitations in Underground Environments

LIDAR-camera fusion has a variety of applications in WIPP-like corridors. It promotes the visual study of geological features and material demarcations, improves flaw detection in walls, ceilings, and support structures, and enables more intuitive comparisons between successive surveys. Fusion, which adds color data to geometrically coherent point clouds, facilitates quantitative deformation analysis and qualitative inspection in massive subterranean tunnels.

Nonetheless, limitations exist. Uniform salt surfaces typically have poor visual texture, limiting the efficiency of image-based analysis. Artificial illumination causes varied exposure and shadowing effects, which may impact color consistency among surveys. Dust and particulate matter reduce optical performance, lowering return strength and image quality. Furthermore, temporal misalignment of sensors may result in spatial aberrations in the integrated point cloud. These restrictions emphasize the significance of solid extrinsic calibration, dependable synchronization, and regular data-gathering processes for reproducible and interpretable outcomes.

2.5.5 Role in Salt Closure Monitoring

In deformation monitoring, geometry is the fundamental measurement variable. Color does not directly improve range precision, but it does help with interpretability and validation. Colorized geometry enables faster visual verification of convergence zones, localized spalling, and surface changes while calculating cross-sectional differences between temporally separated scans.

As a result, this work employs LIDAR-camera fusion as an interpretability layer, together with a geometrically coherent SLAM-derived map. The fusion technique maintains metric precision while boosting situational awareness and facilitating the visualization of long corridor segments.

CHAPTER 3

SYSTEM DESIGN

This chapter describes the hardware and software components of a wearable LIDAR mapping backpack intended for corridor-scale subsurface data collecting. The system was designed to address the following requirements: portability, mechanical rigidity, repeatable geometry acquisition, interoperability with SLAM workflows, and reliable operation in GPS-denied environments. The next sections describe the components used, such as custom mounts, system assembly, validation testing, and communication with the Robot Operating System (ROS2) framework.

3.1 Components Used

The backpack includes four key hardware components: a digital LIDAR sensor, four RGB cameras, an onboard computer for data collection and processing, and a battery pack. Each component was chosen to strike the right balance between geometric integrity, integration flexibility, power consumption, and mechanical robustness.

3.1.1 LIDAR Sensor

The primary geometric sensing component is a *Ouster OS1-64* digital spinning LIDAR. The sensor's rotational scanning architecture provides full 360-degree horizontal coverage and multi-beam vertical resolution. It generates accurately timestamped point clouds over Ethernet, facilitating synchronization and integration inside ROS2.

The sensor was configured to operate at 20 Hz, with 64 vertical channels and 45° horizontal resolution. These values were intended to offer sufficient point den-

sity for corridor cross-section reconstruction while reducing bandwidth and storage requirements during long traversals.

The sensor's digital architecture enables deterministic timing and configurable data output. To ensure correct registration, SLAM-based mapping requires consistent timestamping and a steady scan frequency.

3.1.2 Cameras

The camera subsystem consists of four *ELP-USB500W02M-L21* USB cameras (Figure 3.1) that are rigidly mounted to the LIDAR frame. Each camera operates at 640x480 resolution and 10 frames per second. The cameras offer radiometric context, allowing the point cloud to be colored and subsurface surfaces to be more easily understood.



Figure 3.1: One of four ELP USB cameras.

The camera arrangement was designed to optimize overlap between tunnel ribs and roof regions. To work, projection-based fusion requires at least one camera's field of view to contain 3D points. To reduce blind spots and ensure continuous coverage, an angular direction and spacing were used.

3.1.3 Computer for Pointcloud Processing

All sensing data is acquired and logged using an onboard computer, specifically the *Jetson Orin Nano* shown in Figure 3.2. The machine is running Ubuntu 22.04 with

ROS2 Humble, which contains the LIDAR driver, camera nodes, SLAM pipeline, and rosbag2 recorder.

The key specifications include a *6-core Arm Cortex-A78AE v8.2 64-bit CPU*, NVIDIA Ampere architecture, 8GB of RAM, and support for external NVMe storage. The PC was chosen to provide high-throughput Ethernet communication from the LIDAR while also recording several camera streams with no frame drops.

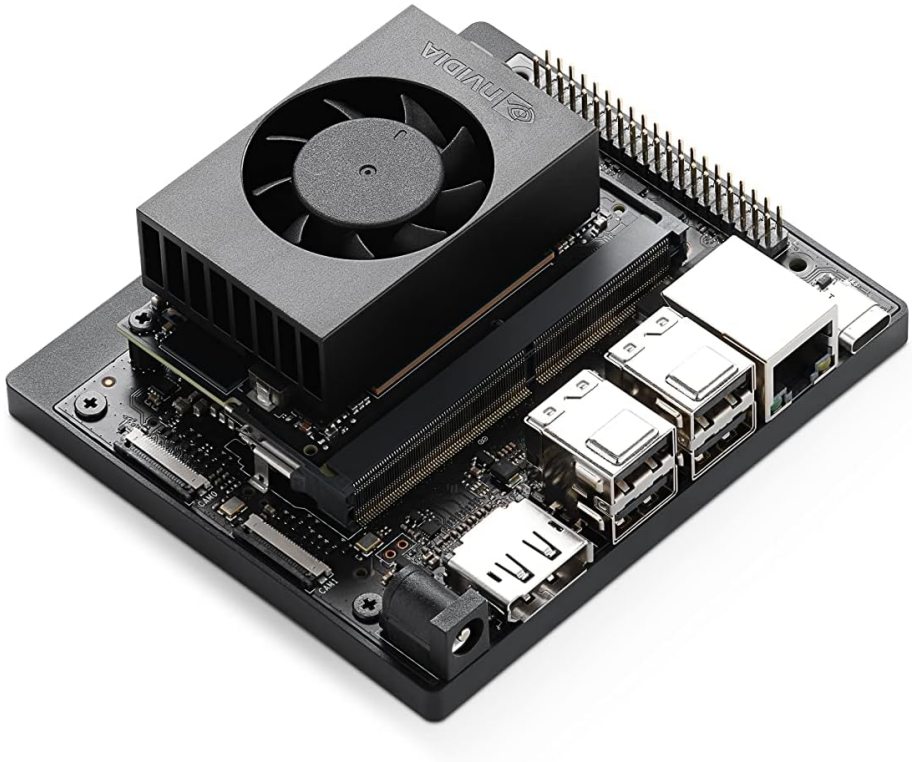


Figure 3.2: Jetson Orin Nano computer used for LIDAR backpack system.

3.1.4 Battery

The system is fueled by a 100-watt *Mophie Powerstation Pro* power bank (Figure 3.3). The pack powers the LIDAR, cameras, and onboard computer. The design goal was to maintain uninterrupted corridor coverage for approximately 2 hours each deployment.

Battery location was regarded as a structural variable because it influences the total center of mass and operator comfort. Keeping mass near to the body lowers oscillatory motion when walking, which improves SLAM stability and scan registration accuracy.



Figure 3.3: Morphie Powerstation Pro battery pack.

3.2 Generative Designed Mounts

Custom mounts were designed using Fusion 360's generative design techniques to assure mechanical stiffness and reduce sensor misalignment. The goal was to de-

sign lightweight, robust components that could tolerate vibrations and loads during underground deployments.

3.2.1 Sensor Package Mount

The sensor package mount keeps the LIDAR and cameras in place while pointing the sensing assembly 45 degrees downward from horizontal. This angular position was chosen to ensure maximum coverage of the tunnel ceiling, ribs, and floor during forward traversal. At 45 degrees, the LIDAR scan plane makes contact with the drift cross-section, capturing both vertical convergence and lateral deformation in a single pass, reducing blind spots that would otherwise occur with strictly horizontal or vertical alignment. To avoid overlap between geometric and radiometric fields of view, the camera array remains in the same orientation.

The generative design technique optimized material distribution based on load scenarios that simulated walking-induced vibration (1.8-2.0 Hz) and incidental impacts. The resulting shape, as illustrated in Figure 3.4, maximizes stiffness-to-weight ratio while staying 3D printable using existing methods. The sensor package mount improves spatial coverage for corridor-scale deformation monitoring by combining angular optimization and structural rigidity.

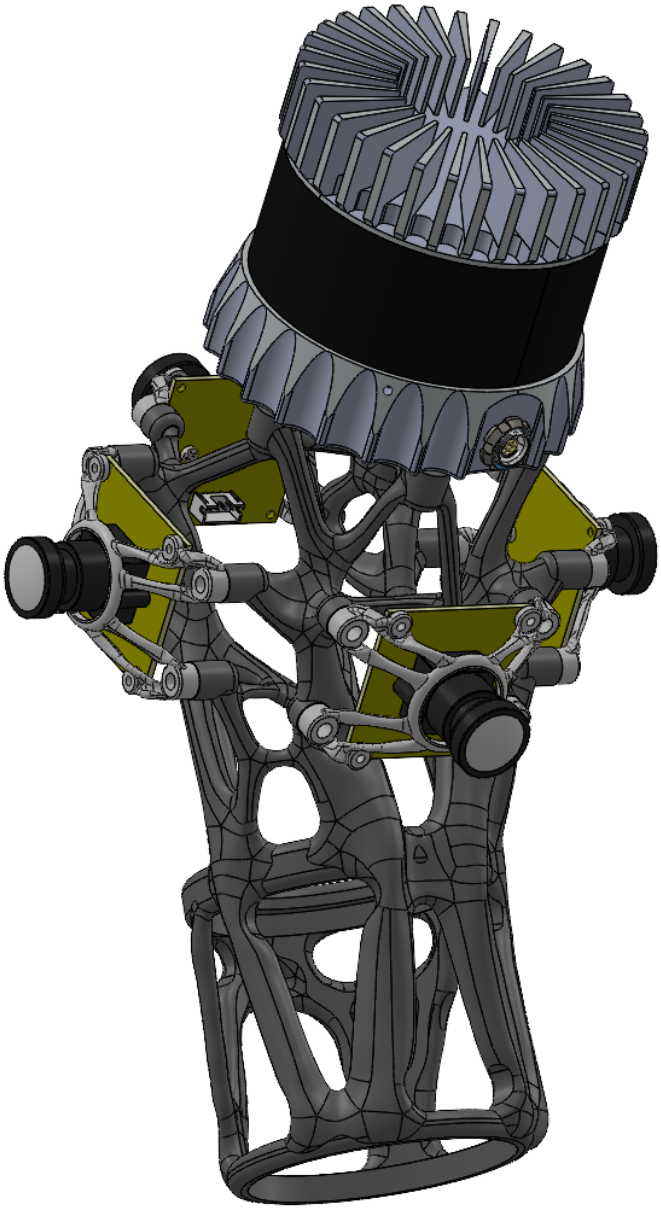


Figure 3.4: CAD model of generatively designed sensor package mount.

The rigid fixation of the sensor package ensures the validity of CAD-derived extrinsic transformations and prevents drift in relative pose between LIDAR and cameras.

3.2.2 Tube Clamp

The tube clamp connects the sensor package to the backpack frame (Figure 3.5). It was designed to withstand rotational slide while being simple to install and remove. The clamp form was modified to effectively distribute clamping pressures along the tube while minimizing stress concentrations that could cause cracking under repeated loading.

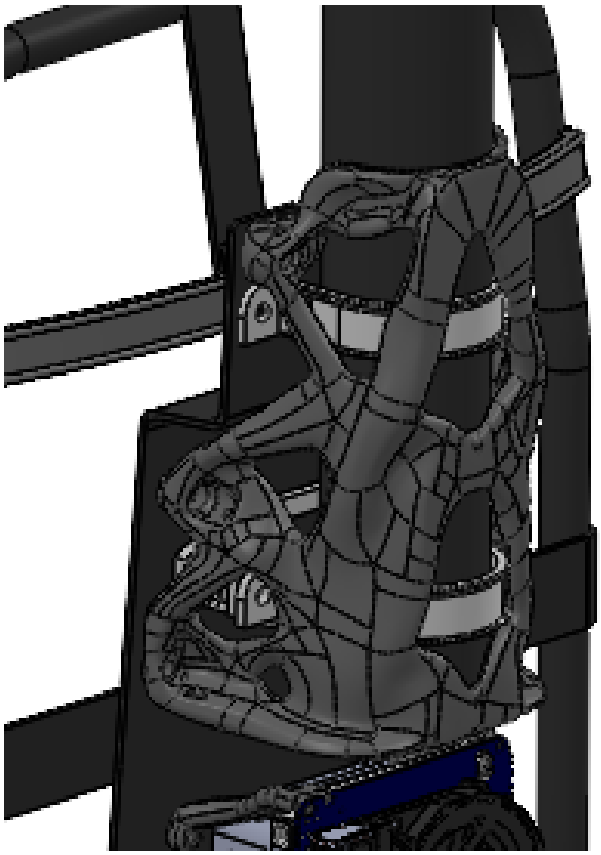


Figure 3.5: Custom tube clamp connecting the sensor mount to the structural frame.

3.2.3 Computer Mount

The computer mount secures the onboard CPU unit inside the backpack container. The mount was designed to isolate high-frequency vibration while maintaining solid

positional stability. The design included enough airflow space to reduce thermal throttling during long recording sessions as depicted in Figure 3.6.

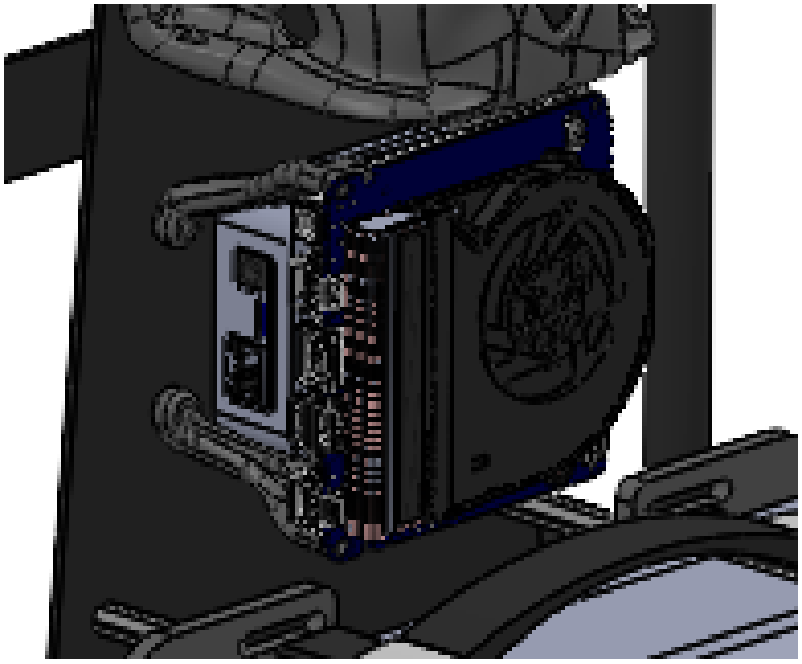


Figure 3.6: Generatively designed computer mounting bracket.

3.2.4 Battery Mount

The battery mount secures the power unit near the operator's center of mass at the bottom of the frame (Figure 3.7). The structural analysis took into account both static load and dynamic movement conditions. The battery attachment bracket is built into the backpack frame for convenient charging and secure retention during underground travel

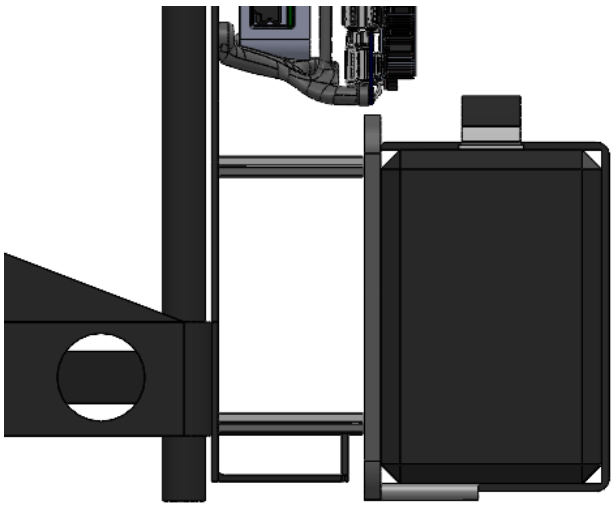


Figure 3.7: Battery mounting bracket integrated into the backpack frame.

3.3 System Assembly

The system was built utilizing a modular approach. The structure was assembled first, followed by the battery and compute enclosure. The sensor package mount was then secured to the frame using the tube clamp, and the LIDAR and cameras were attached to the mount. Ethernet and power cables were routed along shielded pathways to prevent interference with the scan plane and limit the likelihood of snags. The final assembly is shown in Figure 3.8.

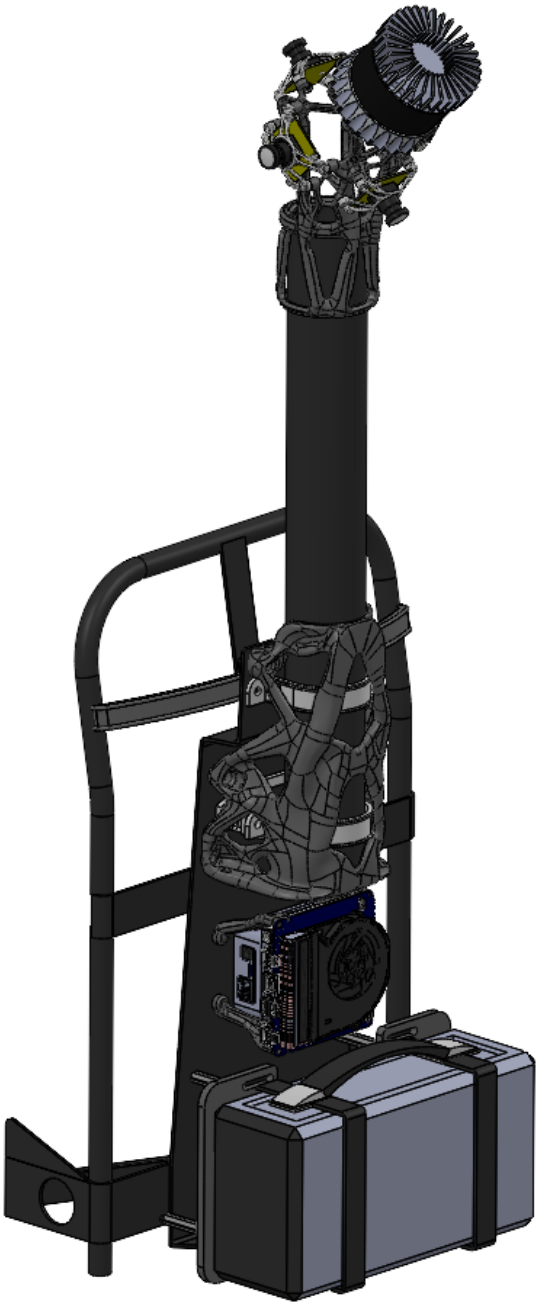


Figure 3.8: Full CAD model of the LIDAR backpack.

Final verification included torque checks on mounting hardware, confirmation of sensor alignment with respect to the CAD reference frame, and validation of secure cable connections. The repeated building process ensures that each deployment starts from the same mechanical baseline.

3.4 Sensor Testing

Before underground demonstration, controlled testing was carried out to guarantee sensor performance and communication stability.

3.4.1 LIDAR IP Configuration

The Ouster LIDAR connects via Ethernet and requires a static IP address to connect to the onboard computer. The computer's network interface was configured using *Netplan* to match the LIDAR's subnet. Ping tests were performed to ensure connectivity and metadata reception from the Ouster driver within ROS2. Proper IP configuration is essential because packet loss or bad network configuration can disrupt data collecting and corrupt rosbag recordings.

3.4.2 Bandwidth Restraints with Cameras

The simultaneous streaming of multiple USB cameras imposed bandwidth limitations on the onboard computer. Testing was conducted to determine the best stable configuration with no dropped frames. Changes included reduced resolution and frame rate, as well as, where available, camera dispersal across multiple USB controllers. These tests ensured that the combined data flow from the LIDAR and

camera array remained within hardware constraints while retaining adequate image quality for projection-based colorization.

3.5 ROS2 Integration

ROS2 offers a middleware architecture for sensing, logging, and mapping components. ROS2 enables modular communication between nodes through publish-subscribe messaging, resulting in time-synchronized data streams. An example of these data streams is illustrated in Figure 3.9.

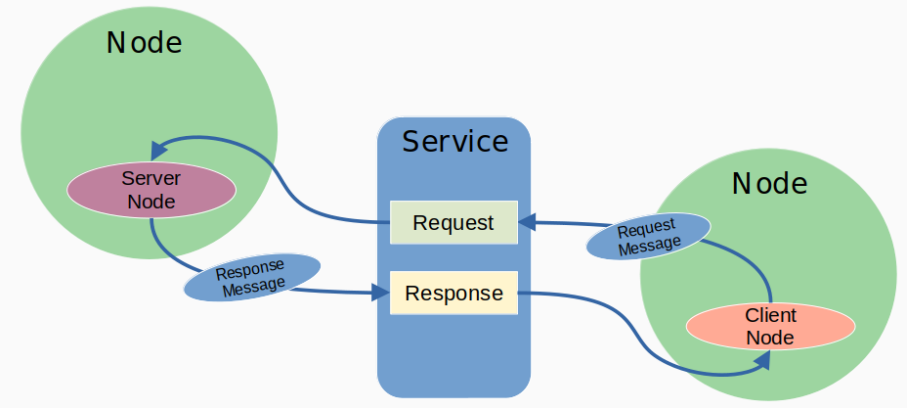


Figure 3.9: Workflow of a ROS2 node.

The backpack system uses ROS2 to control sensor drivers, static transforms, rosbag recording, and SLAM execution. ROS2 supports reproducible launch configurations and identical topic naming across deployments, allowing for offline processing and cross-epoch comparisons.

3.5.1 Ouster Package

The Ouster ROS2 driver generates point clouds, IMU data, and LIDAR metadata. The driver configuration file specifies sensor IP addresses, UDP ports, scan

rates, and resolution parameters. The SLAM process relies heavily on the generated */ouster/points* topic for geometric input.

3.5.2 USB Camera Package

A tailored or customized USB camera ROS2 package broadcasts image streams from all connected cameras. The parameters are device index, resolution, frame rate, and pixel format. The LidarSlam package stores camera subjects, LIDAR data, and static transforms for eventual colorization.

3.5.3 LidarSlam Package

The Kitware LidarSLAM application generates registered maps and estimated trajectories from recorded datasets [41]. The software manages scan registration, drift prevention, and optional loop closure. The generated point clouds and trajectories are exported and compared across survey epochs. In this architecture, SLAM acts as the geometric backbone, transforming raw range data into metrically consistent maps suitable for deformation monitoring. The modular ROS2 integration ensures that sensor drivers, SLAM processing, and fusion components can be interchanged and extended for future system upgrades.

CHAPTER 4

SENSOR FUSION AND COLORIZATION

The sensor fusion architecture intends to augment the reconstructed geometry with radiometric data from RGB cameras while keeping the metrics of LIDAR-based SLAM intact. The fusion technique is deliberately modular. A projection-based augmentation layer adds color, while LIDAR-centric SLAM guarantees that the shapes remain consistent. This disaggregation ensures that deformation measurements are never based on anything except active ranging data. On the other hand, visual information makes it easier to examine and comprehend.

The fusion pipeline is built on the ROS2 middleware framework and consists of four concurrent processes: the Ouster driver, which sends point clouds; the USB camera nodes, which send synchronized image streams; the static transform broadcaster, which defines the rigid transform tree; and the custom colorization node, which converts three-dimensional LIDAR points into image dimensions. The technique's modular design enables live and offline rosbag playback. Data from subsurface demonstrations at WIPP is frequently recorded first and then processed to reduce processing time during collection.

4.1 Transform Architecture and Coordinate Frames

To achieve reliable sensor fusion, coordinate frame definitions must be close together. Misaligning frames, even in a multi-sensor arrangement, can result in projection errors, color misregistration, and spatial distortion in the aggregated data. These flaws, if not managed effectively, can inhibit map comparisons in deformation monitoring applications such as the WIPP salt closure assessment.

The mechanical backbone of the assembly is securely linked to a foundation structure that unites all of the backpack's components. This frame, designated

base_link, serves as the fundamental reference point for the entire transform tree. Figure 4.1 shows how the LIDAR sensor frame and each camera optical frame are coupled to the base frame using fixed rigid-body transformations generated from the sensor package's CAD model. These adjustments expose parts' precise translational offsets and rotational orientations.

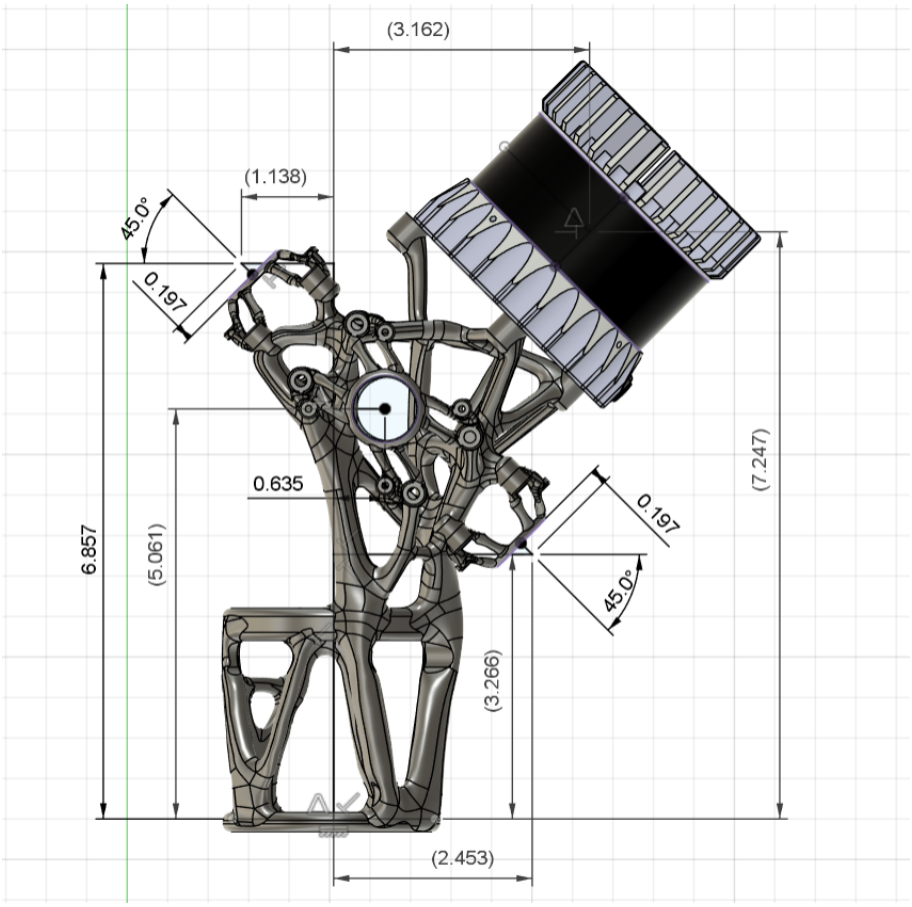


Figure 4.1: Side view of the LIDAR-camera sensor package with dimensions used for static transform calculations.

The ROS2 transform node *static_rig_tf.py* transmits static connections between frames. When this node is started, it sends a sequence of *geometry_msgs/ TransformStamped* messages and establishes a Static Transform Broadcaster. Because

these transformations are static, the TF2 buffer system makes them available at runtime. Consider a location $\mathbf{p}^L$ evaluated within the LIDAR frame:

$$\mathbf{p}^L = \begin{bmatrix} x_L \\ y_L \\ z_L \end{bmatrix}, \tag{4.1}$$

A rigid-body transform of rotation $\mathbf{R}_{BL}$ and translation $\mathbf{t}_{BL}$ determines the transformation to the base frame.

$$\mathbf{p}^B = \mathbf{R}_{BL}\mathbf{p}^L + \mathbf{t}_{BL}. \tag{4.2}$$

The translation vector is directly proportional to the CAD-measured offsets between the LIDAR and base frame origins, but the rotation matrix is generated using quaternion values set in the static transform node. Next, the transformation into the camera's coordinate frame C_i is provided by

$$\mathbf{p}^{C_i} = \mathbf{R}_{C_iB}\mathbf{p}^B + \mathbf{t}_{C_iB}. \tag{4.3}$$

Because the static transform node aligns all of the cameras with the same beginning frame, the entire transform chain is transformed to

$$\mathbf{p}^{C_i} = \mathbf{R}_{C_iB}\left(\mathbf{R}_{BL}\mathbf{p}^L + \mathbf{t}_{BL}\right) + \mathbf{t}_{C_iB}. \tag{4.4}$$

When a colorization node queries TF2, the transformations are immediately merged. When the point cloud message is received, the node wants to know how to transition from the LIDAR frame to an optical frame specific to the camera. TF2 fetches static transforms from its internal buffer by performing correct matrix multiplication and quaternion composition.

Static transforms offer two significant advantages. To begin, it ensures that, as long as the mechanical assembly remains unchanged, the geometry is reproducible between deployments. Additionally, sensors do not alter with time because there is no online extrinsic estimation. The stiffness of the resulting mount adds to geometric consistency.

The necessity to preserve rigidity stems from structural and mathematical factors. Random mechanical shifts $\delta\mathbf{t}$ and tiny rotational disturbances $\delta\boldsymbol{\theta}$ alter the effective extrinsic transform. At a certain distance from the sensor, even minor angular misalignment yields an approximation of the positional error at point R.

$$\Delta x \approx R\,\delta\boldsymbol{\theta}, \tag{4.5}$$

As the range rises, so does this in a straight line. Even little variations in angle can cause centimeter-level projection inaccuracies in deep subterranean tunnels when the surface is several meters away from the sensor. This level of sensitivity enables the extraction of extrinsics from CAD geometry while isolating the mount from walking-induced vibrations.

Figure 4.2 illustrates ROS2's transform hierarchy. The core connecting frame ensures the system's stability. This layer was the foundation for the LIDAR frame as well as the optical frames of all cameras. As a result, a star-shaped topology emerged, simplifying chaining and making it easier to find transforms.

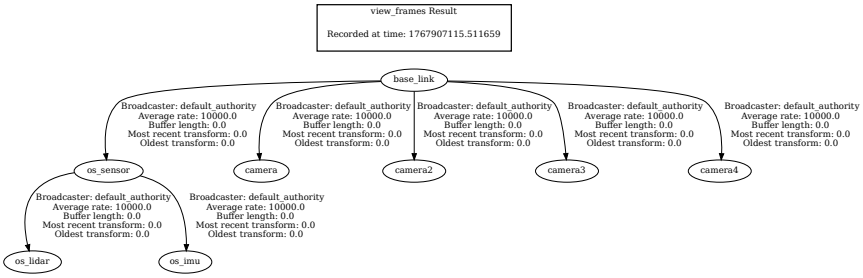


Figure 4.2: Transform (TF) tree of the LIDAR-camera nodal network running on ROS2.

This approach ensures that all subsequent operations, including color assignment, SLAM integration, and projection, use a common and universally determined coordinate system. Because deformation monitoring is based on comparing maps, transform consistency is critical. If there is a persistent error in the static transform requirements, the fused datasets will be distorted, obscuring true physical closure signals.

By identifying extrinsic elements from observed CAD geometry, passing them via ROS2 static transforms, and retaining mechanical stiffness during deployment, the system builds a predictable geometric backbone for all subsequent mapping and fusing processes.

4.2 Intrinsic Calibration and Camera Model

This design regards intrinsic calibration as bypassing the abstract camera model and proceeding directly to the colorization node. Every camera delivers a *sensor_msgs/CameraInfo* message including the intrinsic matrix $\mathbf{K}$, distortion coefficients $\mathbf{D}$, and image resolution parameters. The colorization node monitors the calibration settings for each camera individually by listening for these signals.

The projection model is constructed using the *CameraInfo* data structure, instead of hardcoding intrinsic values, which the node acquires dynamically. This ensures that the projection parameters match those in the actual calibration files used by the USB camera drivers.

For each camera C_i, the intrinsic matrix is extracted as

$$\mathbf{K}_i = \begin{bmatrix} f_{x,i} & 0 & c_{x,i} \\ 0 & f_{y,i} & c_{y,i} \\ 0 & 0 & 1 \end{bmatrix}, \tag{4.6}$$

where $f_{x,i}$ and $f_{y,i}$ are focal lengths expressed in pixel units, and $(c_{x,i}, c_{y,i})$ represents the principal point. In the implementation, these parameters are accessed directly from the flattened 3x3 matrix contained in the ROS2 message:

```
fx = camera_info.k[0]
fy = camera_info.k[4]
cx = camera_info.k[2]
cy = camera_info.k[5]
```

By directly indexing, we can ensure that the projection model matches the camera calibration. After LIDAR points are converted into camera coordinate frames, projections use perspective division. To acquire normalized image coordinates of a point $\mathbf{p}^C = (x_C, y_C, z_C)$, follow the procedures below:

$$x_n = \frac{x_C}{z_C}, \quad y_n = \frac{y_C}{z_C}. \tag{4.7}$$

Pixel coordinates are then calculated explicitly within the node using

$$u = f_x x_n + c_x, \quad v = f_y y_n + c_y. \tag{4.8}$$

In code form, this appears as:

```
u = int((fx * x_cam / z_cam) + cx)
v = int((fy * y_cam / z_cam) + cy)
```

Before projecting pixels, the node ensures that they are within valid picture coordinates by exhaustively inspecting their borders. Points with $z_C \leq 0$ are instantly eliminated to prevent them from being projected outside the camera plane. Prior to RGB sampling, the image's width and height are additionally validated against the pixel indices.

The *CameraInfo* message also offers information on the lens's picture distortion. The current methodology corrects distortion in input images before projection utilizing OpenCV-based rectification methods. The intrinsic matrix will be perfectly aligned with the undistorted coordinate system, and the projection model will work with rectified image frames.

```
dist_coeffs = camera_info.d
```

Color alignment accuracy is directly proportional to intrinsic calibration precision, as each camera performs its own projection. The projected pixels change in a regular manner due to tiny mistakes in focal length or main point, causing the colorized cloud's edges to misalign. Differentiating the projection equations with regard to f_x and c_x allows you to identify the extent to which the projection is affected by intrinsic error. From this, we can conclude that the pixel error rises with depth and radial distance from the focus point.

The technique combines runtime fusion and calibration data by utilizing explicit projection equations within the node and dynamically collecting intrinsic parameters from ROS2 camera topics. This method ensures that colorization is constant across all deployments by matching calibration files and projection logic.

4.3 Projection-Based Colorization Algorithm

The study foregoes the tightly coupled visual-LIDAR SLAM strategy in favor of a projection-based fusion method. The speed with which projection-based colorization is completed has little effect on the accuracy of geometric forecasts. Large-scale mapping procedures, where geometry is crucial and color assists comprehension, commonly utilize equivalent methods [42, 43].

The custom ROS2 node is updated by the picture topic for each camera and the LIDAR point cloud subject. In response to a point cloud message, the node fetches the relevant static transform from TF2, converts each point into a potential camera frame, projects the point using the intrinsic model, and samples the related RGB value. The result is delivered as a *PointCloud2* message with colors.

The projection loop is depicted here in simplified pseudocode.

```
for point in pointcloud:
    p_base = transform_lidar_to_base(point)
    for cam in cameras:
        p_cam = transform_base_to_camera(p_base, cam)
        if p_cam.z > 0:
            u, v = project_to_image(p_cam, cam.K)
            if pixel_in_bounds(u, v):
                rgb = image[v, u]
                assign_color(point, rgb)
                break
```

To guarantee the geometric backbone remains stable prior to radiometric assignment, this modular method allows for colorization based on replay after SLAM optimization.

4.4 Integration with SLAM and Map Consistency

LidarSLAM is based on Kitware's framework [38]. It provides optimal trajectories and registered point clouds. SLAM establishes a stable global reference frame by utilizing scan registration and posture graph optimization to correct drift [44, 31].

Colorization is used once the geometric uniformity has been established. Deformation measurements produced by comparing maps are thus assured to be independent of camera projection errors. We consider two mappings, $\mathcal{M}_1$ and $\mathcal{M}_2$, separated by time. Deformation is judged in this manner:

$$\Delta(\mathbf{x}) = d\big(\mathcal{M}_1(\mathbf{x}), \mathcal{M}_2(\mathbf{x})\big), \tag{4.9}$$

$d(\cdot)$ represents the distance between two surfaces or clouds. In closure analysis, radiometric fusion is more concerned with understanding than measuring because color is not a component of the measure.

4.5 Validation Assessment and Interpretive Impact

The Applied Research Center (ARC) Robotics Laboratory provided a controlled indoor environment for testing the projection-based colorization technology before it was used underground. The goal of this testing phase was not to detect reprojection faults down to the millimeter scale, but rather to ensure geometric consistency, create suitable transform chaining, and see if radiometric augmentation improved interpretation.

Some of the components evaluated were the Ouster driver, camera nodes, static transform broadcaster, and custom colorization node. Rosbag recordings were played back to test and troubleshoot them in a controlled environment. Manual inspection of the fused output confirmed, from various angles, that the projected color was

consistent with geometric edges and surface boundaries. The projection's accuracy was tested utilizing visual aids such as structural beams, high-contrast objects like lab furniture edges, and wall-floor interfaces. The final colorized map is shown in Figure 4.3.

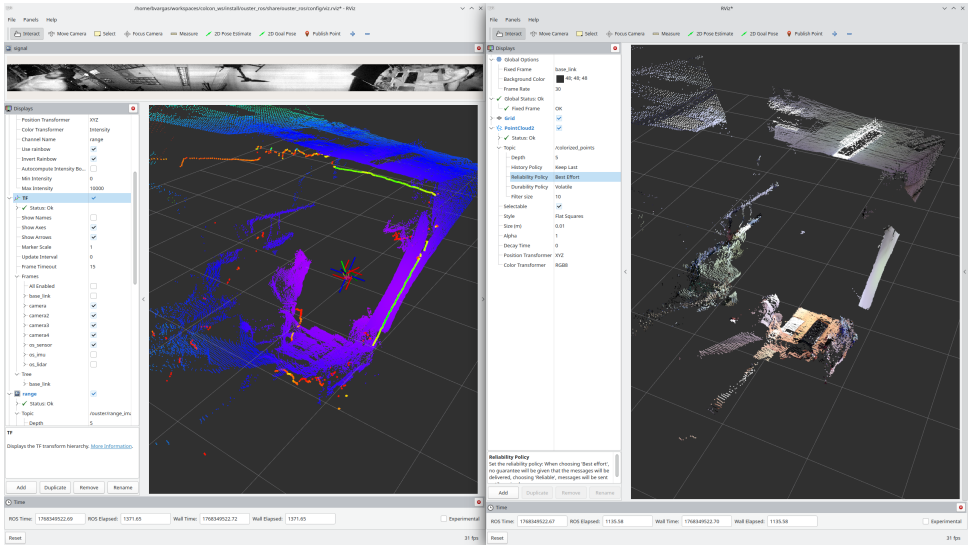


Figure 4.3: Colorized point cloud generated in FIU's Applied Research Center Robotics Laboratory.

The colorful point cloud helps you better understand the surface. While the SLAM-generated map maintains its fundamental geometry, the addition of RGB information allows for faster distinction of materials, structural components, and surface discontinuities. This enrichment significantly simplifies things for the general public while maintaining the metric quality used in deformation analysis.

When moving slowly, the projection was consistent with very few ghosting artifacts. The previously established relationship between time and distance, $\Delta x \approx v\Delta t$, was validated by the slight misalignment seen at greater speeds. Due to the use of timestamped LIDAR and picture streams in projection, any remaining synchronization latency appears as slight color shifts in fast-changing sceneries. Slowing down causes these artifacts to shrink, but they keep their original shape.

In the colorization node's current implementation, each point and camera candidate is projected sequentially. For dense point clouds, this indicates it's slow. Despite the fact that it was not built for real-time throughput, the node performs well and has geometric consistency. At this stage, the goal was to prove the concept rather than to improve the calculations. The laboratory results demonstrating successful hand colorization justify the mathematical projection model, transform architecture, and ROS2 integration.

The most important result of this testing phase for deformation monitoring is evidence that radiometric augmentation does not disrupt the geometric backbone that SLAM develops. The primary purpose for comparing maps remains LIDAR-derived spatial coordinates. Color adds a layer of interpretation that improves situational awareness, accelerates the process of identifying convergence zones, and increases the value of exported maps for engineering evaluation.

The separation of geometry and radiometry is purposeful. By employing colorization after SLAM optimization, the technique retains the metric consistency required for quantitative salt closure analysis. By design, this architecture removes the potential of sensor fusion bias as the cause of any observed displacement between surveys undertaken at different times.

Future enhancements could include implementing methods to speed up the projection loop, utilizing the Jetson platform's GPU capabilities to run multiple tasks at the same time, optimizing extrinsic parameters by minimizing reprojection error, and assigning colors based on occlusion to avoid camera conflicts. Surface characterization in subterranean areas with little color contrast may benefit from the addition of LIDAR intensity data.

Laboratory validation shows that the projection-based fusion method simplifies things while keeping geometric correctness. This capacity improves the wearable LI-

DAR backpack system's effectiveness for corridor-scale monitoring at WIPP, where both quantitative deformation measurement and qualitative inspection are used in practice.

CHAPTER 5

POINT CLOUD CHANGE DETECTION

The primary purpose of this thesis is to demonstrate how repeatable 3D mapping can detect spatial changes in underground tunnels. Salt creep, which manifests as the gradual but persistent convergence of drift floors, ribs, and roofs over time, is a physical phenomenon of interest at WIPP. Instead of inferring closure from sparse instrumentation, repeated 3D maps can be aligned into a stable reference frame and compared using surface-to-surface distance measurements. This will enable the spatial quantification of closure. This chapter presents controlled validation studies that corroborate the sensitivity of the post-processing strategy and lays out the methodical road from raw point clouds to quantified deformation fields.

5.1 Point Cloud Post-Processing

Point cloud post-processing transforms raw scan data into metrically consistent datasets that can be compared across epochs. The method starts with cleaning and exporting raw scans in a common coordinate frame to remove noise and outliers. After filtering, datasets undergo rigid registration and geometric differencing. All deformation analysis was performed using CloudCompare, an open-source software platform. It is specifically designed for 3D metrology and surface comparison.

Change detection in CloudCompare is typically performed using cloud-to-cloud (C2C) or cloud-to-mesh (C2M) distance metrics. For two point clouds $\mathcal{M}_1$ and $\mathcal{M}_2$, the C2C method computes, for each point $\mathbf{p}_i \in \mathcal{M}_1$, the minimum Euclidean distance to the nearest neighbor $\mathbf{q}_j \in \mathcal{M}_2$:

$$d_i = \min_{\mathbf{q}_j \in \mathcal{M}_2} \|\mathbf{p}_i - \mathbf{q}_j\| . \tag{5.1}$$

The resulting scalar field $\{d_i\}$ shows a color map over the surface, indicating the size of spatial displacement. Doing this calculation for each site yields a continuous deformation field rather than a solitary convergence value. Within the WIPP framework, this allows for a full cross-sectional examination of drift geometry, as well as the simultaneous evaluation of roof, rib, and floor behavior.

To test the sensitivity of this approach, a controlled experiment was conducted in an Applied Research Center corridor with a Trimble X7 terrestrial laser scanner. These studies were designed to isolate the capabilities of the post-processing software under high-precision scanning conditions, rather than employing the wearable LIDAR backpack. Sheets of standard printer paper were placed in 2 millimeter increments ranging from 0 to 10 millimeters. Figure 5.1 shows the setup used for controlled millimeter-scale validation. Each increment was scanned separately, with the same acquisition settings.



Figure 5.1: Trimble X7 scanning setup used for controlled millimeter-scale validation.

The baseline scan $\mathcal{M}_0$, which did not include a paper stack, served as a reference. We utilized an ICP-based registration technique in CloudCompare to reduce translation and rotation offsets from successive scans $\mathcal{M}_2, \mathcal{M}_4, \ldots, \mathcal{M}_{10}$ and firmly

aligned them to $\mathcal{M}_0$. The next stage was to compute the distance between clouds following the alignment.

The quantity of interest in this verification was the relative normal displacement of the paper and desk surfaces in the vertical (Z) direction. In addition to the total Euclidean distance, CloudCompare also delivers signed distance components ($\Delta X, \Delta Y, \Delta Z$)". The observed values match ΔZ, which reflects displacement normal to the desk plane. The numbers are negative because the measurement was taken from the raised paper surface to the desk reference. As a result, the sign represents the direction rather than the magnitude of the mistake.

The measured vertical displacements were:

- 2 mm test: -0.002000 m
- 4 mm test: -0.004200 m
- 6 mm test: -0.005837 m
- 8 mm test: -0.008037 m
- 10 mm test: -0.010234 m

Figures 5.2 through 5.6 illustrate the deviation maps for each increment relative to baseline.

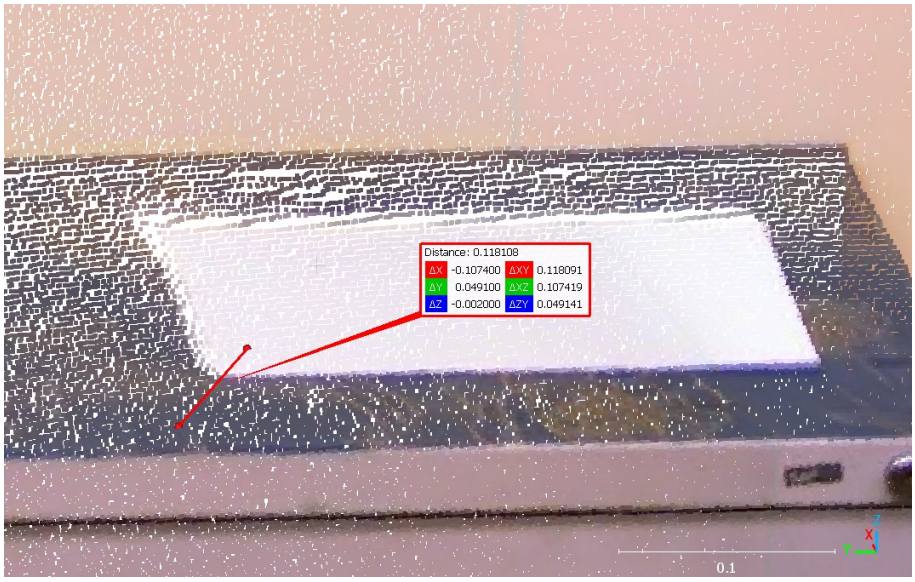


Figure 5.2: Cloud-to-cloud distance map showing approximately 2 mm displacement relative to baseline.

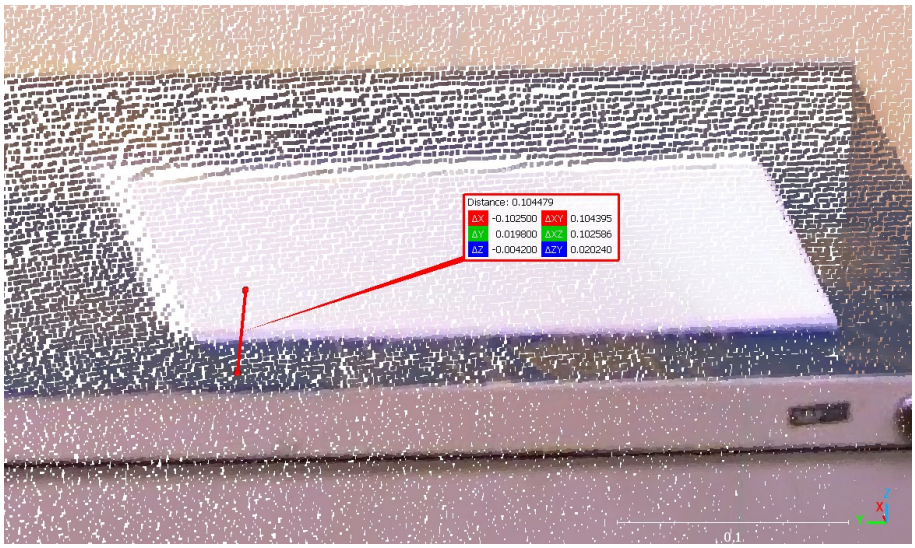


Figure 5.3: Cloud-to-cloud distance map showing approximately 4 mm displacement relative to baseline.

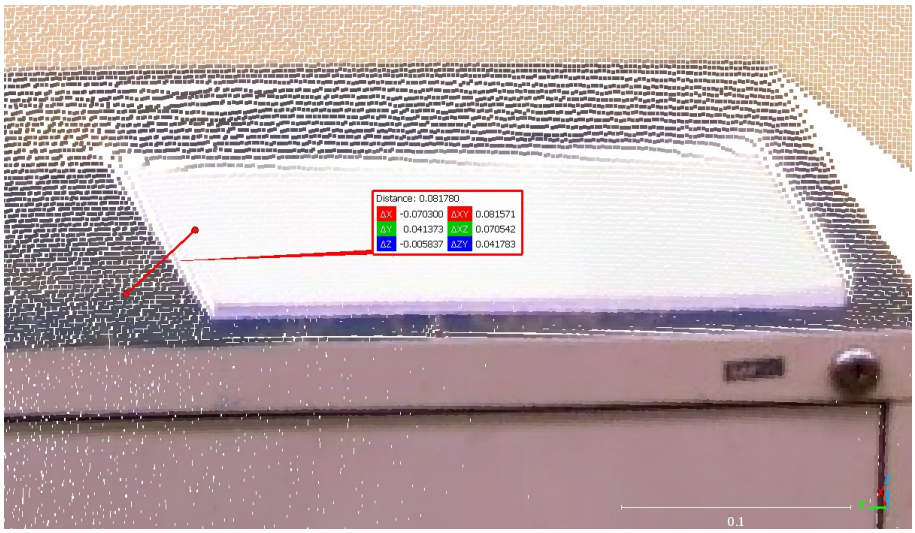


Figure 5.4: Cloud-to-cloud distance map showing approximately 6 mm displacement relative to baseline.

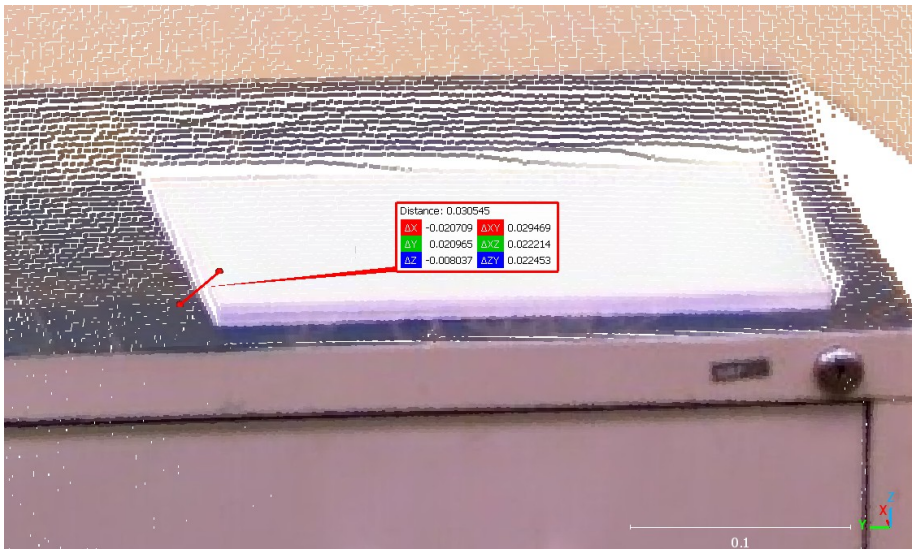


Figure 5.5: Cloud-to-cloud distance map showing approximately 8 mm displacement relative to baseline.

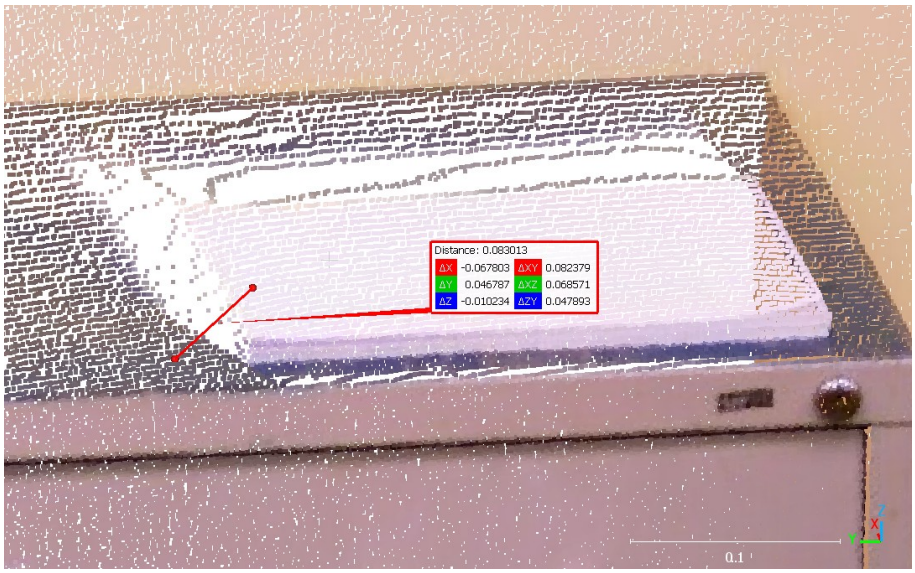


Figure 5.6: Cloud-to-cloud distance map showing approximately 10 mm displacement relative to baseline.

The results reveal that the estimated geometric deviation and the forced physical displacement are strongly correlated. The 2mm test provided a precise reading of -0.002000 m. With a 0.2-millimeter difference from nominal, the 4-millimeter example yielded -0.004200 m. Cases measuring 6 millimeters, 8 millimeters, and 10 millimeters produced results of -0.005837, -0.008037, and -0.010234 millimeters, respectively. Across all trials in the 10 mm test, the maximum absolute departure from the nominal thickness was approximately 0.234 mm.

Under controlled conditions, these findings reveal that the forced displacement and measured Z-direction deviation are within millimeters of one another. There is no nonlinear distortion injected into the post-processing workflow throughout the examined range, as evidenced by the near-linear scaling of the imposed height and measured displacement.

The typical stated rate of salt convergence at WIPP ranges from millimeters to centimeters each year. Steady-state roof-to-floor convergence rates in operating drifts can range from 50 to 150 mm/year, depending on excavation geometry, stress

redistribution, and panel age [45]. As long as registration uncertainty and measurement noise are kept under control, the sub-millimeter detection capability can detect deformation magnitudes required to monitor the closure of repository-scale systems.

5.2 Point Cloud Registration

The ability to perform reliable change detection at WIPP depends fundamentally on consistent point cloud registration across both space and time. Registration serves two related purposes in this research. First, it enables the stitching of sequential scans into a single, spatially continuous corridor-scale map. Second, it allows time-separated surveys to be aligned within a common reference frame so that geometric differences represent physical deformation rather than pose error.

Rigid registration between two point clouds $\mathcal{M}_1$ and $\mathcal{M}_2$ seeks the transformation $(\mathbf{R}, \mathbf{t})$ that minimizes

$$\min_{\mathbf{R},\mathbf{t}} \sum_i \|\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i\|^2 , \tag{5.2}$$

where $\mathbf{p}_i$ and $\mathbf{q}_i$ are corresponding points in the two datasets. In the Trimble X7 validation experiment, rigid-body translation and rotation were removed using Iterative Closest Point (ICP) before differencing, and this alignment was performed directly in CloudCompare.

Continuous registration occurs during acquisition of the wearable LIDAR backpack using LIDAR-centric SLAM. Scan matching and loop closure, when possible, refine the projected course as the operator walks down a corridor, with each scan aligned to a growing local map. Instead of obtaining scan fragments, our method recreates the entire traversed corridor using consistent metrics.

This stitching capacity is critical for WIPP-scale installations in which drifts might exceed hundreds of meters. This approach avoids the need for individual tripod surveys by combining many traversals into a single continuous geometric model, allowing geometry to be captured at the corridor size while in motion.

A field test was performed at the Port of Miami tunnel to determine the reliability of large-scale registration in a monotonous tube environment (Figure 5.7). The LIDAR backpack is secured to the sunroof of a car traveling at around fifteen miles per hour. With the addition of dynamic motion, this experiment aimed to recreate the visually repetitive geometry of a subterranean drift. This traversal covered around 171 meters of corridor.

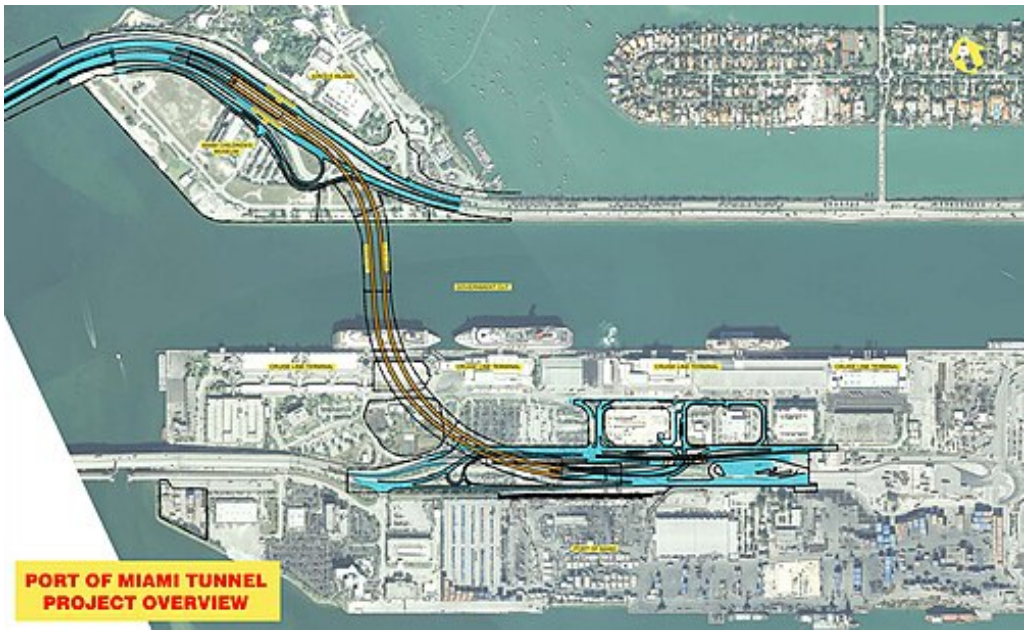


Figure 5.7: Overview of the Port of Miami Tunnel in Miami, FL.

To demonstrate intermediate SLAM consistency over extended distance, Figure 5.8 displays the reconstructed point cloud of the split tunnel segment before final alignment. After the section was registered and optimized, Figure 5.9 shows the complete 171 m segment fused into a continuous model.

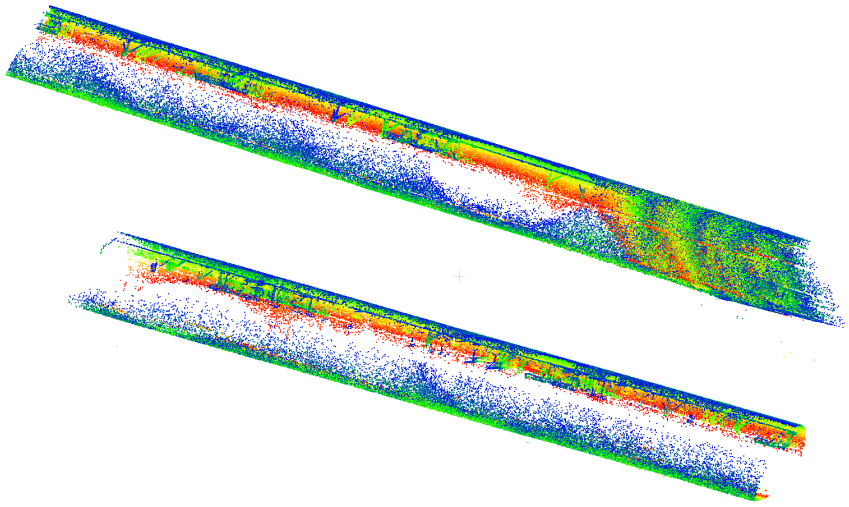


Figure 5.8: Port of Miami tunnel reconstruction shown in two registered halves prior to final stitching.

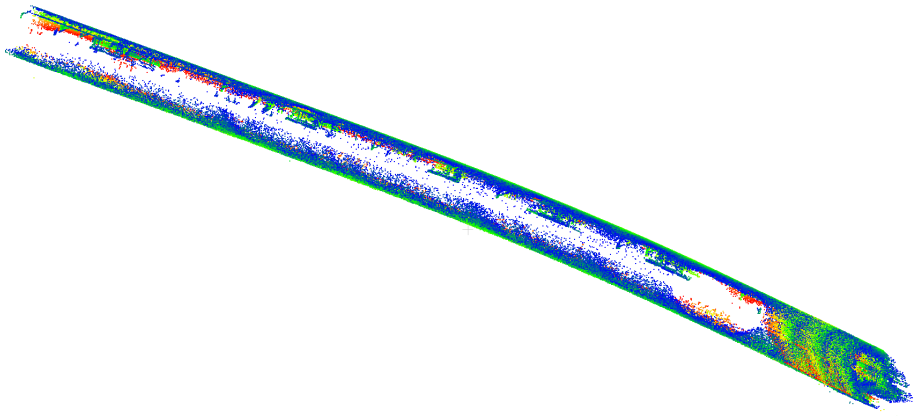


Figure 5.9: Full 171 m section of the Port of Miami tunnel stitched into a single continuous point cloud using SLAM-based registration.

Extending tunnel-scale mapping in dynamic environments is doable without catastrophic drift buildup, as demonstrated by the reconstructed model. Confirming

that scan registration and trajectory estimation are robust even in feature-repetitive circumstances, the geometry remains consistent throughout the collected length. This observation has significant significance for WIPP, as prolonged drifts exhibit similar geometric monotony.

The accuracy of change detection depends on both scanner precision and registration quality. If σ_s represents measurement noise and σ_r represents residual registration uncertainty, the effective detection threshold σ_d can be approximated as

$$\sigma_d \approx \sqrt{\sigma_s^2 + \sigma_r^2}. \tag{5.3}$$

Maintaining σ_r below the expected magnitude of salt closure ensures that computed deviation fields reflect true physical deformation rather than accumulated trajectory error.

After obtaining corridor-scale maps, repeated surveys convert the system into a geometric time machine. Each iteration generates a spatial model $\mathcal{M}(t)$ that represents the drift geometry at time t. The deformation may be computed by comparing two surveys performed at times t_1 and t_2, using the formula

$$\Delta(\mathbf{x}, t_1, t_2) = d\big(\mathcal{M}(t_1), \mathcal{M}(t_2)\big), \tag{5.4}$$

where $d(\cdot)$ represents the distance between clouds or surfaces. The internal stitching of the surveys into a single continuous frame allows for differentiation across entire drift extents rather than just limited local locations.

This method allows you to generate a permanent master map from a series of WIPP scans. By comparing new surveys to this worldwide norm, we may determine whether there has been any change over time. Unlike fixed convergence devices, which measure displacement between discrete anchor points, this approach

captures continuous geometric progression throughout entire drift cross-sections and longitudinal distances.

Adding color to these stitched corridor models makes them more understandable. Engineers can employ radiometric context to link convergence zones to structural elements, installed hardware, or geologic boundaries, whereas geometry is used to determine the extent of displacement. Without affecting the metric basis of analysis, deviation maps layered over colored geometry provide rapid spatial context for observed change.

The potential operational basis for a future WIPP demonstration may involve combining multi-epoch registration and SLAM-based corridor stitching. Rather than quantifying salt creep as separate sites, sequential scans can be incorporated into unified corridor models to directly compare repeated traversals.

CHAPTER 6

CONCLUSION

6.1 Summary of Objectives and Contributions

The primary aim of this thesis was to illustrate the viability of employing repeated three-dimensional mapping to identify deformation caused by salt creep in underground passageways at WIPP. The primary concept was that geometric differencing may identify quantifiable closure tendencies, contingent upon the generation of spatially consistent maps across many temporal epochs and their alignment within a stable coordinate framework.

This thesis presents a comprehensive end-to-end workflow, encompassing the collection of mobile LIDAR data and the identification of changes between point clouds. The pipeline incorporates hardware design, data acquisition using ROS2, SLAM with LIDAR, projection-based colorization, and subsequent analysis in CloudCompare. The objective of each design step was to maintain the metric's validity and enable more spatial comparisons.

The LIDAR backpack is compliant with WIPP operational criteria. The created mounts maintain the close interactions of all the sensors. Simultaneous logging with both LIDAR and cameras in a GPS-denied environment is now feasible due to the Jetson-based computing architecture. The system may be effortlessly relocated, quickly configured, and utilized repeatedly for corridor surveys.

Rather than perceiving SLAM as a method to augment accuracy, individuals regarded it as a tool to improve visibility. LIDAR-centric SLAM offers the necessary geometric framework for comparing findings across various time intervals by sustaining a consistent trajectory and reducing drift. The principal reason for alterations

in deformation estimates, as opposed to physical changes, would be algorithmic aberrations due to the lack of continuous registration.

We employed a Trimble X7 terrestrial laser scanner to perform controlled millimeter-scale experiments for evaluating the change-detection toolchain. The findings of the paper-stack experiment indicated that displacements analogous to the documented rates of salt convergence can be identified using the post-processing pipeline. This illustrates that the analytical method can identify shape alterations that are crucial for repository-level monitoring.

Finally, metric independence is preserved by the modular LIDAR-camera fusion architecture, which incorporates radiometric context into geometry. While colorization improves reading and increases environmental awareness, it does not influence the deformation metric. This division asserts that active ranging data is the exclusive foundation for quantitative results.

6.2 Implications for Salt Closure Monitoring at WIPP

Currently, WIPP is meticulously monitoring geotechnical conditions with instrumented, point-based measurements, including extensometers and convergence meters. Despite their ease of installation and maintenance, these systems have limited coverage and produce a significant amount of relevant data.

The solution presented in this thesis enhances existing technologies by ensuring spatial consistency over entire drifts. Dispersed deformation patterns are undetectable with a limited number of measurements; nevertheless, mobile LIDAR mapping can identify them. By repeatedly creating these maps, one may witness convergence across the entire surface and track temporal changes. This methodology offers color-coded maps for enhanced visibility, quick deployment without permanent

installation, consistent geometry for quantitative differentiation, and corridor-scale geographical coverage.

- Corridor-scale spatial coverage,
- Rapid deployment without permanent installation,
- Repeatable geometry suitable for quantitative differencing,
- Visual interpretability through colorized maps.

Transitioning from sparse point measurements to distributed surface fields enhances the monitoring paradigm from discrete convergence readings to comprehensive surface deformation analysis. This facilitates improved ground control planning, prioritization of maintenance, and evaluation of repository engineering tasks.

6.3 Limitations

We must acknowledge specific constraints. Instead of employing the wearable backpack, a terrestrial scanner was utilized for millimeter-scale validation. Nonetheless, this does not yet assess the field accuracy of the mobile system; it merely delineates the functionalities of the post-processing technique.

Furthermore, SLAM drift and registration quality continue to affect cross-epoch alignment. Estimates of deformation magnitude can be influenced by minor pose variations in extensive passages with limited characteristics. Verifying registrations and diligently resolving outstanding issues are essential.

Thirdly, the analysis of dense point clouds is protracted due to the sequential operation of the existing projection-based colorization node across points and cameras. While the implementation is functional, it requires optimization for large-scale application.

A fourth issue is that, although the technical architecture is presented in this thesis, longitudinal datasets collected directly at WIPP over prolonged durations are still absent. Ongoing surveys performed under actual repository settings are essential for comprehensive operational validation.

6.4 Future Work

Future study should concentrate on computational optimization, accurate characterization of wearable platforms, and subsurface demonstration at WIPP.

6.4.1 Optimization of the Colorization Node

The current implementation processes each point sequentially and performs per-point TF lookups, transformations, projections, and RGB packing operations [**?**]. This architecture, while functional, results in increased processing time that escalates linearly with the number of points and cameras. It is advisable to focus optimization in three principal domains.

1. Elimination of Redundant Per-Point Operations For each prospective point and camera, the existing system performs TF lookups and rigid transformations within the innermost loop. It is advisable to request transformations solely once for each camera and point cloud, thereafter converting them into homogeneous matrices, as the extrinsic relationship between the LIDAR and each camera remains constant throughout a callback. The complete point cloud can be altered by replacing the per-point *do_transform_point* procedures with vectorized matrix multiplication. This change alone substantially decreases the overhead of loops at the Python level and access to TensorFlow buffers.

2. Vectorization and GPU Acceleration Currently, each point sequentially receives RGB assignment and projection. NumPy arrays facilitate the execution of vectorized operations on point clouds, enabling tasks such as perspective division, pixel projection, and batch processing. The intrinsic parallelism of transformation and projection renders these processes advantageous for CUDA acceleration on the Jetson Orin Nano. Enhancing the speed of dense Ouster scans and approaching real-time performance can be accomplished by GPU-based matrix multiplication and filtering.

3. Intelligent Camera Selection and Calibration Refinement The node now identifies the initial legitimate projection following the examination of each camera separately. Geometric pre-filtering based on the camera's orientation is employed to minimize superfluous projection efforts. Choosing the camera with the minimal incidence angle, instead of the initial successful match, can enhance color constancy. The geometric framework employed for deformation analysis can be maintained, while the reduction of reprojection error facilitates enhanced edge alignment. This is because it can somewhat reduce the mechanical tolerances of extrinsic parameters.

By delineating geometry and color augmentation, these enhancements facilitate system scalability, enhance radiometric consistency, and diminish resource requirements.

Algorithmic acceleration will enhance offline processing speed, potentially enabling near real-time viewing during deployment. Maintaining the geometric precision necessary for deformation monitoring is achievable without subsampling when handling denser point clouds, which is an additional advantage of minimizing performance overhead.

Enhancing extrinsic calibration to reduce reprojection error could improve color alignment accuracy, facilitating comprehension while preserving a clear distinction between geometry and radiometry. These improvements will transform the node from a proof-of-concept into a scalable fusion module appropriate for prolonged subterranean application.

6.4.2 Demonstration at WIPP

The primary objective of this research is to apply it within the WIPP underground environment. The subsequent elements must be incorporated into the demonstration procedure:

- Baseline corridor scanning,
- Repeat surveys over defined time intervals,
- Cross-epoch map alignment,
- Quantitative deformation field extraction,
- Comparison against existing convergence instrumentation.

Integrating LIDAR deformation fields with sensor observations yields corroboration from many sources. The reliability of both types of monitoring would improve if discrete convergence meters and distributed surface measurements were consistent.

6.5 Closing Remarks

Creating reproducible three-dimensional maps facilitates the identification of alterations in subterranean conduits, as evidenced in this thesis. A unified framework for

spatial salt closure monitoring has been established by the integration of LIDAR-centric SLAM, rigid mechanical design, ROS2-based integration, and cloud-to-cloud differencing.

Over time, each survey serves as a geometric record of the repository's state. These recordings demonstrate deformation as measurable spatial fields instead of distinct point values when analyzed over time. Should the wearable LIDAR backpack system undergo further refinement and field testing, it may augment our understanding of salt creep mechanisms, facilitate operational monitoring at WIPP, and integrate with current geotechnical instruments.

BIBLIOGRAPHY

[1] U.S. Department of Energy, Office of Environmental Management. *Waste Isolation Pilot Plant (WIPP)*. U.S. Department of Energy, 2024. Retrieved from https://www.energy.gov/em/waste-isolation-pilot-plant-wipp.

[2] U.S. Department of Energy, Office of Environmental Management. *Waste Isolation Pilot Plant (WIPP)*. U.S. Department of Energy, 2024. Retrieved from https://www.energy.gov/em/waste-isolation-pilot-plant-wipp.

[3] U.S. Department of Energy. *History and Timeline of the Waste Isolation Pilot Plant*. U.S. Department of Energy, 2024. Retrieved from https://wipp.energy.gov/historytimeline.asp.

[4] Nuclear Active. Rapid seven-fold, never-been-observed-at-wipp increases in salt convergence in panel 7 where waste is planned for disposal, 2021. Retrieved from https://nuclearactive.org/rapid-seven-fold-never-been-observed-at-wipp-increases-in-salt-convergence-in-panel-7-where-waste-is-planned-for-disposal/.

[5] William J. Broad. Nuclear waste solution seen in desert salt beds. *The New York Times*, February 2014. Published February 9, 2014. Retrieved from https://www.nytimes.com/2014/02/10/science/earth/nuclear-waste-solution-seen-in-desert-salt-beds.html.

[6] U.S. Department of Energy. *Waste Isolation Pilot Plant News Release*. U.S. Department of Energy, 2024. News release, January 24, 2024. Retrieved from https://wipp.energy.gov/wipp$_n ews_2$0240124.*asp*.

[7] U.S. Department of Energy. Waste isolation pilot plant news release, October 2021.

[8] U.S. Department of Energy. *Waste Isolation Pilot Plant Site Overview*. U.S. Department of Energy, 2024. Retrieved from https://wipp.energy.gov/wipp-site.asp.

[9] U.S. Department of Energy. *Geology and Repository Design of the Waste Isolation Pilot Plant*. U.S. Department of Energy, 2024. Retrieved from https://wipp.energy.gov/geology.asp.

[10] U.S. Environmental Protection Agency. *EPA's Role at the Waste Isolation Pilot Plant*. U.S. Environmental Protection Agency, 2023. Retrieved from https://www.epa.gov/radiation/epas-role-waste-isolation-pilot-plant-wipp.

[11] National Research Council. Creep behavior of wipp salt. In *The Waste Isolation Pilot Plant: A Potential Solution for the Disposal of Transuranic Waste*, pages 123–141. The National Academies Press, 1996. doi:10.17226/5269.

[12] National Research Council. Underground operations at the waste isolation pilot plant. In *Safety in the Underground Construction and Operation of the Exploratory Studies Facility*, pages 90–100. The National Academies Press, 1995. Ground control and deformation monitoring.

[13] Sandia National Laboratories. Isolation in salt: Repository performance at wipp, 2025. Technical overview of salt creep and repository closure.

[14] D.E. Munson and P.R. Dawson. Salt creep, closure, and performance assessment at the waste isolation pilot plant. *International Journal of Rock Mechanics and Mining Sciences*, 21(4):217–228, 1984.

[15] Martin Štroner et al. Scanning the underground: Comparison of the accuracies of slam and static laser scanners in a mine tunnel. *Measurement*, 2025. ScienceDirect article page: https://www.sciencedirect.com/science/article/pii/S0263224124017603.

[16] J. Kang et al. A coal mine tunnel deformation detection method using point cloud data. *Sensors*, 2024. Open-access via PubMed Central: https://pmc.ncbi.nlm.nih.gov/articles/PMC11013994/.

[17] Mateusz Janiszewski, Masoud Torkan, Lauri Uotinen, and Mikael Rinne. Rapid photogrammetry with a 360-degree camera for tunnel mapping. *Remote Sensing*, 14(21):5494, 2022. https://www.mdpi.com/2072-4292/14/21/5494.

[18] Nuclear Waste Partnership LLC. *WIPP Geotechnical Engineering Program Plan*, 2017. Effective Date: 04/12/2017.

[19] U.S. Department of Energy. *Title 40 CFR Part 191 Subparts B and C Compliance Recertification Application 2019: Appendix MON (Monitoring Programs)*, 2019. March 26, 2019. Retrieved from https://wipp.energy.gov/library/CRA/CRA

[20] Waste Isolation Pilot Plant (U.S. Department of Energy). Underground aerial survey conducted at wipp, June 2024. Published June 5, 2024. Retrieved from https://wipp.energy.gov/wipp$_n ews_2$0240605.*asp*.

[21] NOAA Office for Coastal Management. *An Introduction to Lidar Technology, Data, and Applications*, 2012. Retrieved from https://coast.noaa.gov/data/digitalcoast/pdf/lidar-101.pdf.

[22] Erika Podest. *The Fundamentals of LiDAR*. NASA Applied Remote Sensing Training Program, 2021. Retrieved from https://appliedsciences.nasa.gov/sites/default/files/2021-03/SIF$_L IDAR_P odest_F inal.pdf$.

[23] Texas Instruments. *LIDAR Pulsed Time of Flight Reference Design*, 2018. Retrieved from https://www.ti.com/lit/pdf/tiducm1.

[24] Thomas Skoog et al. Fmcw lidar: Scaling to the chip-level and improving phase noise with coherent detection. Technical report, University of California, Berkeley, EECS, 2019. Retrieved from https://www2.eecs.berkeley.edu/Pubs/TechRpts/2019/EECS-2019-148.pdf.

[25] Hamamatsu Photonics. *Measuring Distance with Light*, 2021. Retrieved from https://hub.hamamatsu.com/us/en/application-notes/automotive/measuring-distance-with-light.html.

[26] SPIE. *Lidar Range Equation*, n.d. Retrieved from https://spie.org/samples/FG36.pdf.

[27] Ocean Optics Web Book. The lidar equation, 2021. Retrieved from https://www.oceanopticsbook.info/view/radiative-transfer-theory/level-2/the-lidar-equation.

[28] Velodyne Lidar, Inc. *VLP-16 User Manual and Datasheet*, 2020. Technical specifications and system architecture overview.

[29] Trimble Inc. *Trimble X-Series 3D Laser Scanners Technical Specifications*, 2023. Product technical documentation.

[30] Ouster, Inc. *OS-Series Digital Lidar Sensor Specifications*, 2023. Product datasheet and integration documentation.

[31] Wolfgang Hess, Damon Kohler, Holger Rapp, and Daniel Andor. Real-time loop closure in 2d lidar slam. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 1271–1278, 2016.

[32] Kitware, Inc. Loop closure is ready to use in lidarview!, January 2023. Kitware blog post.

[33] Ji Zhang and Sanjiv Singh. Loam: Lidar odometry and mapping in real-time. *Robotics: Science and Systems (RSS)*, 2014. Often cited as the original LOAM work.

[34] Tixiao Shan and Brendan Englot. Lego-loam: Lightweight and ground-optimized lidar odometry and mapping on variable terrain. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018.

[35] Tixiao Shan, Brendan Englot, Carlo Ratti, and Daniela Rus. Lio-sam: Tightly-coupled lidar inertial odometry via smoothing and mapping. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020.

[36] Wei Xu and Fu Zhang. Fast-lio2: Fast direct lidar-inertial odometry. *arXiv*, 2022.

[37] Stéphane Vajna et al. Kiss-icp: In defense of point-to-point icp for lidar odometry. *arXiv*, 2023.

[38] Kitware, Inc. Slam (kitware lidar slam), 2019. GitLab repository.

[39] Kitware, Inc. lidar_slam (ros2 wrapping) readme, 2025. Project documentation.

[40] Kitware, Inc. How to slam with lidarview, 2025. Project documentation.

[41] Kitware, Inc. Lidarslam, 2024. Open-source SLAM package for LiDAR-based mapping. Accessed February 2026.

[42] Brian Curless and Marc Levoy. A volumetric method for building complex models from range images. *Proceedings of SIGGRAPH*, 1996.

[43] Qian-Yi Zhou, Jaesik Park, and Vladlen Koltun. Open3d: A modern library for 3d data processing. *arXiv preprint arXiv:1801.09847*, 2018.

[44] Giorgio Grisetti, Rainer Kümmerle, Cyrill Stachniss, and Wolfram Burgard. A tutorial on graph-based slam. *IEEE Intelligent Transportation Systems Magazine*, 2010.

[45] U.S.-German Workshop on Salt Repository Research, Design, and Operation. Proceedings of the 10th u.s.-german workshop on salt repository research, design, and operation. In *Proceedings of the 10th U.S.-German Workshop on Salt Repository Research, Design, and Operation*. Sandia National Laboratories, 2019. Reports steady-state roof-to-floor convergence rates at WIPP of approximately 50–150 mm/year.